Land at South Marston and Rowborough
Swindon

Transport Assessment

Swindon EV Consortium

Brookbanks

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Executive Summary

Brookbanks Consulting (BCL) is appointed by the Swindon Eastern Villages Consortium to produce a Transport Assessment and Travel Plan to accompany a planning application for a residential development at South Marston and Rowborough to the east of Swindon.

The proposed development will comprise up to 2380 new houses, a primary school and supporting uses including a local centre and mixed use area incorporating leisure & community facilities, low scale employment opportunities and open space provision. This represents a comprehensive first phase in the development of the Eastern Villages.

As part of evidence prepared in support of the Swindon Eastern Villages development proposals, BCL has worked in conjunction with Swindon Borough Council (SBC) and the Highways Agency (HA) representatives to work toward agreement on all highways matters, including trip generation forecasts, traffic modelling methodology and results, together with the production of a package of mitigation measures to enhance the highway network and accessibility of the area while mitigating for transport impacts of the development proposals.

The process followed has been used by BCL as the basis for this Transport Assessment, with the key agreements applied to the development proposals.

In relation to transport policy, the proposed development accords with national, regional and local transport and land-use policies. These support Swindon Borough Council's aim to promote sustainable development that seeks to ensure wider travel choices in the form of the most sustainable forms of transport and thus reduce the use of the private car particularly for single person use. The proposals seek to address positively the principles set out in relation to South Marston in the Draft South Marston SPD and take account of the guidance also in the Draft Eastern Villages SPD.

A number of vehicular accesses are proposed into the existing highway network, being from a compact roundabout on Thornhill Road, with access also from several locations off Old Vicarage Lane and from a signalised junction on the A420 towards the eastern extent of the site. These accesses are consistent with the Local Plan evidence base documents that have been tested independently and again within this document. In accordance with the Draft South Marston and Eastern Villages SPD no access is taken off Nightingale or Rowborough Lanes.

The western access point off Thornhill Road will provide access to the South Marston residential development. The proposed accesses off Old Vicarage Lane will provide a secondary point of access serving the expanded South Marston. The development area at Rowborough will be served by accesses from / to Old Vicarage Lane and via the eastern access under the railway.. The phasing of the connections back to the A420, including the timing of the delivery of the rail under bridge at the eastern end of the site, will be a matter for discussion with SBC.

Arrangements to enhance public transport provision are provided together with new and enhanced footways and cycleways in the area which will reduce the need to travel by private car.

Pedestrian and cycle connections to the neighbouring South Marston and Sainsbury's development and the public transport interchange at Gablecross are proposed to be enhanced and potential effects of severance caused by the A420 is proposed to be reduced through the provision of upgraded junctions with crossing facilities.

A much improved bus service operating at a 20 minute frequency will connect the development to the town centre, rail station and bus interchange at Fleming Way.

It is agreed that a key component of the transport mitigation measures is the implementation of a comprehensive Travel Plan (TP), which is of a scale sufficient to warrant a standalone document accompanying this Transport Assessment. The Travel Plan proposes a target for the site which will reduce the mode share by single occupancy vehicles by 10%.
The final components of the package of transport mitigation measures are the provision of highway improvements either directly by the developer or through developer contributions at various locations on the A420 and A419 and in South Marston.

The potential impacts on the highway network have been tested through the Swindon area wide Saturn traffic model together with the A419 / A420 Paramics traffic model. This has resulted in the identification of a range of offsite interventions that will delivered by the development. This included:

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road.

- **New Access onto A420:** New signal controlled junction at the east end of phase one to provide access from the A420
- **Police HQ Signals:** - Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire
- **Gablecross Roundabout:** Localised widening on approach arms
- **White Hart interchange:** Partial signalisation of the eastern side of the interchange

This represents a comprehensive package of interventions delivered through, and arising from, the development. The detailed traffic modelling completed for this TA has also considered future development resulting from the wider growth proposals to the east of Swindon, as identified in the Local Plan. As would be expected, this work has identified further interventions that will be necessary to deliver the wider growth. However, the assessments demonstrates that delivery of this Phase I element of the growth does not prejudice the delivery of the subsequent phases. Such further interventions that have been anticipated in the Eastern Villages SPD for instance and include a diamond interchange at White Hart and the signalisation of the Greenbridge roundabout. The SPD envisages the phased implementation of wider transport improvements in accordance with the phased implementation of the eastern villages development.

Overall, the development provides mitigation in relation to the transport networks and aims to minimise travel through the implementation of the Travel Plan to sufficiently minimise the impact of the development on the highway network. Good connectivity is afforded to the surrounding urban area and accessibility to a range of key services, facilities and opportunities has been provided.

Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston. The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston. Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is discouraged from travelling towards South Marston. This proposed junction arrangement was also considered to be effective in avoiding traffic passing through South Marston in the technical studies conducted by JMP as background to the draft South Marston SPD.

A connection through South Marston and connections at Rowborough will be provided, the routes of which have been identified by the masterplan to connect to the external public highway. Access from Rowborough onto Old Vicarage Lane is proposed at two points. By this means the traffic modelling work confirms that traffic generated from within Rowborough towards Old Vicarage Lane will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road. As such the transport strategy route provides an equitable and effective route choice. The proposed link between Old Vicarage Lane and Thornhill Road is therefore demonstrated not to perform a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

The potential impact of the proposals for South Marston has been fully appraised, including Pound Corner. This assessment indicates that the delivery of these proposals will not materially increase the flows through the village or impede access to the A420 for existing residents.

Furthermore in relation to the highway assessment, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included. Whilst the flow of traffic on those southern sections of Thornhill Road and Old Vicarage Lane down to the A420 inevitably increase, the traffic flows between the access points and the existing village generally reduce in the morning and am peaks, following the Phase 1 development. This is likely to reflect the arrangement of the junctions to discourage traffic movements to the north but also the choice of some existing users of the links to remain on other sections of the network rather than to pass through South Marston as a result of the additional junctions that would need to be navigated. For these and other reasons there is no material increase in traffic anticipated through Pound Corner.

Notwithstanding, the additional traffic generated by the development, with the improvements proposed to the Gablecross roundabout and the junction of Old Vicarage Lane, the level of queuing at Gablecross roundabout on the Thornhill Road arm is forecast to reduce compared to the no development and no improvement position and the queuing on Old Vicarage Lane will be limited and acceptable.

The impact on South Marston will therefore be limited by:

- Discouraging the use of new highway links north of the A420 and south of South Marston being used in preference to the A420.

- Discouraging vehicles using South Marston village.

Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the emerging Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

In summary, the proposals demonstrate that a well-considered approach to developing transport and highways proposals for the first phase of development at Swindon Eastern Villages is able to ensure the proposals are able to meet national, regional and local policy and guidance while meeting with the identified aspirations of the growth area.

1 Introduction

1.1 Brookbanks Consulting Limited (BCL) is appointed by the Swindon Eastern Village Consortium (SEVC) to complete a Transportation Assessment (TA) and a Travel Plan (TP) for a proposed mixed use development on land to the east of Swindon. The proposed site is part of the wider Eastern Villages (EV) growth area that is supported by Swindon Borough Council planning policies and the adopted Local Plan.

1.2 Development to the east of Swindon has been mooted over many years, supported through various iterations of the Borough Council's Local Development Framework. Initially, the Plan policies identified that circa 12,000 dwellings could be built on land to the east of Swindon. More recently this has been reduced to circa 8,000 dwellings. The identified land parcel that forms the basis of this planning application represents Phase I of the Eastern Villages and lies to the north of the A420, near South Marston. The future phases of development will complete the Eastern Villages and these will be subject to their own planning application.

1.3 The objective of the study is review the Phase I application in detail and to demonstrate that the development proposals are acceptable from a transportation and highways viewpoint. This report will also review the future phases at a high level to demonstrate that the wider growth is not prejudiced by delivery of Phase I.

2 Background Information

Location

2.1 The proposed development lies to the east of Swindon. The site is generally bound to the south by the Swindon – Paddington rail line, beyond which lies the A420. Open fields, generally in agricultural production bound the site to the east and north. The Village of South Marston lies to the north west of the site with Thornhill Road forming the western boundary of the site. Old Vicarage Lane bisects the land, forming the boundary between the South Marston and Rowborough proposals as recognised in the Local Plan documents.

2.2 The land is largely historically undeveloped. The site location and boundary are shown indicatively on Figure 2a

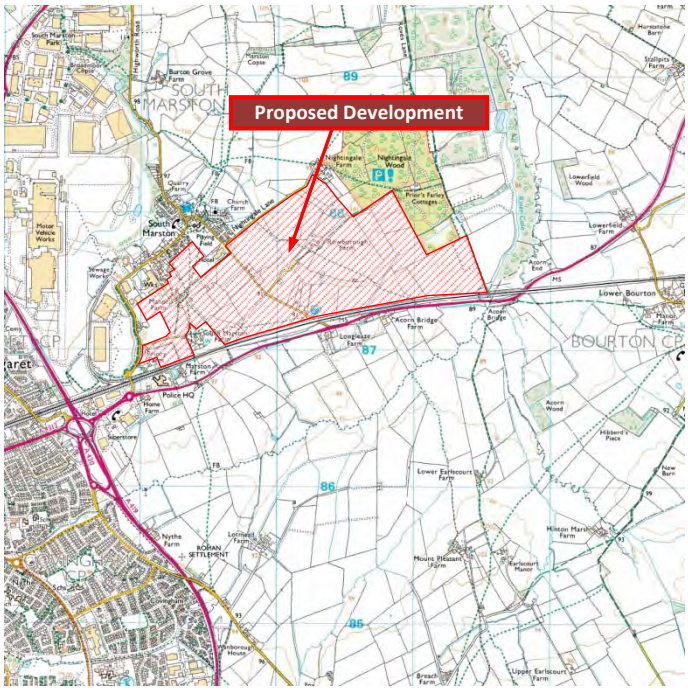

Figure 2a: Site location

Scheme Proposals

2.3 The Phase I Proposed Development comprises of two parts, being the growth of South Marston and to the east, a development of a new neighbouring community at Rowborough. Each area will have its own distinct identity and a strong landscape and green infrastructure framework will define and shape the development of South Marston and Rowborough as individual settlements. Particular emphasis has been placed on protecting South Marston from potential undesirable effects of the growth.

2.4 The application provides for some 580 dwellings at South Marston and a development that will build on the existing village character and improve links to the surrounding countryside. The layout comprises of a series of residential land parcels developed on a connected network of routes, which integrate with the existing village. The existing school, beyond the application site, is proposed to be extended in accordance with the South Marston Supplementary Planning Document (SPD). Provision is made for recreation and community facilities for the expanded village. A large area of open land between the expended village and railway line is to maintain and enhance identity of the village. Measures are promoted to limit traffic within the existing and proposed extended village in line with previously agreed principles.

2.5 Some 1,800 dwellings are proposed at Rowborough to form one of the larger new villages within the EV allocation. A new primary school and mixed use local centre(s) are also proposed. The stream valley area provides a strategic landscape corridor, which will remain undeveloped, punctuated by retained hedgerows and trees. The landscape corridor will provide attractive informal open and/or amenity space and reinforce separation between South Marston and Rowborough.

2.6 The Development Framework Parameter Plan, attached in Appendix A, sets out built development components, areas of formal and informal open space and the alignment of the primary strategic transport routes running through the application site. Key layout principles are embedded within the Parameter Plan and illustrated through the illustrative master plan:

- The development will be comprised of walkable residential neighbourhoods around distinct character areas. Each neighbourhood benefits from access to key areas of formal and informal open space.

- An accessible and central 'Neighbourhood Hub', created through co-location of community facilities, will form the focus of activity for Rowborough and be accessibly located on the main through street. A mixed use centre is proposed to the south eastern area of Rowborough, again adjoining the main through street. The South Marston SPD sets out proposals to enhance local facilities in South Marston.

- Opportunities for strategic footpath / cycle linkages into the wider network will be exploited, connecting adjoining locations within the wider Swindon Eastern Villages to the application site.

- Legibility of the development will be ensured by developing a positive relationship between buildings, streets and spaces, with buildings fronting onto and providing opportunities for overlooking and surveillance. In addition, ensuring there is a strong definition of public (streets, opens space) and private spaces (back gardens, private driveways) making the layout legible and safe.

- A variety of landscape and development responses to the existing character of the site, including provision of a robust landscape framework including a network of formal and informal open spaces, and the creation of sports pitches located in convenient and accessible locations. At South Marston pitches are located to provide recreation opportunities for the wider community.

- The development will establish a rich and diverse green infrastructure strategy incorporating existing trees and hedgerows and connecting parks, play areas, formal sports pitches, and wildlife habitats.

2.7 The planning application considers a total of 2,380 new dwellings. The detailed traffic modelling, summarised in this document, assumes 2,550 dwellings, to represent a worst case (highest impact) scenario.

2.8 Future phases of the growth have been tested and are identified as Phase IA and Phase II in accordance with profiles of growth agreed during pre application consultations and as set out in the Eastern Villages SPD.

- Phase I represents this planning application.

- Phase IA provides for Phase I plus 40,000 ha of B1/ B2 and B8 land uses to the south of the A420 together with an expansion of the Sainsbury's supermarket expansion from 7,350sqm to 10,000sqm in line with criteria agreed during consultations.

- Phase II considers the latest Local Plan allocation, in that to deliver:

  > An additional 6,000 residential dwellings south of the A420
  > retail floorspace located in a network of local centres
  > Eight form entry secondary school
  > Eight forms of entry of primary school provision
  > Doctors surgery, Dentist and Pharmacy at the District Centre

2.9 Design of the transport and highway proposals is considerate to the specific concerns of the local South Marston community. The access strategy has been considered to specifically limit the increase in traffic through South Marston. The selection of a roundabout junction on Thornhill Road provides a direct link into the development to reduce any need to travel through South Marston. Proposals for Old Vicarage Lane include the alteration of the priority such that traffic generated by the development is encouraged through the development. Access from Rowborough is proposed onto Old Vicarage Lane at two points. By this means traffic generated from within Rowborough will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road. Access under the railway is also proposed to the east.

2.10 As development progresses a connection through South Marston and connections at Rowborough will be provided the routes of which have been identified by the masterplan to connect to the external public highway. As such the transport strategy route provides an equitable and effective route choice for this development. The proposed link between Old Vicarage Lane and Thornhill Road is not intended to perform a southern bypass role for the South Marston area. Indeed the link has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

2.11 The potential impact of the proposals for South Marston has been fully appraised, including Pound Corner. This assessment indicates that the delivery of these proposals will not materially increase the flows through the village or impede access to the A420 for existing residents. .

2.12 Furthermore in relation to the highway assessment, the Thornhill Road, Old Vicarage Lane and the new access have all been assessed. The results indicate that the junctions will operate satisfactorily when the development has been included.

2.13 The impact on South Marston will be limited by:

- Discouraging the use of new highway links north of the A420 and south of South Marston being used in preference to the A420.
- Discouraging vehicles using South Marston village

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2.14 Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the emerging Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

Transport Assessment Consultations

2.15 In preparation for the planning application submission, detailed discussions have been held with both Swindon Borough Council (SBC) and representatives for the Highways Agency (HA) over an extended period of time. These discussions have intensified over the last nine months as the planning application has been prepared. Various aspects covering the delivery of the scheme have been discussed with fundamental principles addressed and agreed. This has led to a coordinated approach to agree key aspects of scheme delivery.

2.16 This process has resulted in working agreements across various several areas of the assessment, including the following key areas:

- Use of the Swindon Wide Saturn traffic model for the high level overview
- Use of the A419 / A420 Paramics traffic model
- Location and the form of the Phase I access points
- Methodology to be adopted in order to review the Phase IA and 2
- Trip generation rates to be applied
- Travel Plan principles

Transport Assessment Structure

2.17 The TA covers all the issues required by the DfT publication 'Guidance on Transport Assessment'. This document is intended to assist transport practitioners to determine if a transport assessment is required and if so, the level and scope of that assessment. This document specifically requires the following matters to be addressed:

- Reducing the need to travel, especially by car
- Sustainable accessibility
- Dealing with residual trips
- Mitigation measures

2.18 The report incorporates appropriate text that reflects the agreed matters and the remainder of the report is structured in the following way:

Chapter 3: National and Local Policy Background

This chapter reviews both National and Local Policy documentation to demonstrate that this site is supported for residential use.

Chapter 4: Existing Transport Conditions

This chapter details the site location in relation to the public transport, walking, cycling networks, together with the national and local road network.

Chapter 5: Development Proposals
This Chapter reviews the development proposals and details the proposed access arrangements.

Chapter 6: Development Impact Appraisal
The chapter assesses the development in relation to Accessibility, Safety, Economy, Environment and Integration.

Chapter 7: Travel Plan
This chapter provides details on the Travel Plan that has been drafted to support this development including any measures that can be utilised to achieve the agreed modal shift targets.

Chapter 8: Development Traffic Generation
The Chapter provides details on the expected number of trips generated by this site and the methodology on how they are to be distributed within the local road network.

Chapter 9: Local Road Network Review – Journey Time
This chapter indicates the results of the assessment on journey time as predicted by the Paramics traffic model.

Chapter 10: Local Road Network Review – Link Flow Assessment
This chapter indicates the results of the assessment on link flows as predicted by the Paramics traffic model.

Chapter 11: Local Road Network Review – Junction Assessment
This chapter indicates the results of the assessment on queuing at key junctions within the road network as predicted by the Paramics traffic.

Chapter 12: Limitations
This chapter defines the limitations on the above conclusions based on the accuracy of information received.

3 National And Local Policy Background

National Policy

3.1 The Government White Paper on Transport of 2004 identified the need for a transport network that can meet the challenges of a growing economy and the increasing demand for travel but can also achieve environmental objectives. This requires a transport network that provides a more reliable and freer flowing service while making walking and cycling real alternatives for local trips. There is a clear recognition that the location and nature of development affects the amount and method of travel; and the pattern of development is itself influenced by transport infrastructure and transport policies.

3.2 **PPG13:** The Department of Communities and Local Government, Planning Policy Guidance (PPG) Note PPG13: Transport historically provided advice on how local authorities should integrate transport and land use planning. The planning policy made a contribution towards the goal of improving urban quality and vitality and to achieving a healthy rural economy and viable rural communities.

3.3 As a result, the integration of land use and transport policies became fundamental and development plan preparation led this process by aiming to reduce the need to travel, by:

- Influencing the location of different types of development relative to transport provision (and vice versa)
- Fostering forms of development which encourage walking, cycling and public transport use.

3.4 **NPPF:** PPG13 was superseded when the Department of Communities and Local Government published the National Planning Policy Framework (NPPF) in March 2012. This was produced to support reforms of the planning system and to

promote sustainable growth. The NPPF sets out a framework within which Councils can produce development plans which reflect the needs and priorities of those communities.

3.5 Paragraph 30 within the NPPF encourages solutions that support reductions in greenhouse gas emissions and reduce congestion. Local planning authorities should therefore support a pattern of development which, where reasonable to do so, facilitates the use of sustainable modes of transport.

3.6 Paragraph 32 identifies that all developments generating significant amounts of movement should be supported by a Transport Statement or Transport Assessment. Plans and decisions should take account of whether:

- The opportunities for sustainable transport modes have been taken up depending on the nature and location of the site, to reduce the need for major transport infrastructure
- Safe and suitable access to the site can be achieved for all people
- Improvements can be undertaken within the transport network that cost effectively limit the significant impacts of the development. Development should only be prevented or refused on transport grounds where the residual cumulative impacts of development are severe

3.7 Paragraph 35 within NPPF identifies that plans should protect and exploit opportunities for the use of sustainable transport modes for the movement of goods or people. Therefore developments should be designed where practical to:

- Accommodate the efficient delivery of goods and supplies
- Give priority to pedestrian and cycle movements and have access to high quality public transport facilities
- Create safe and secure layouts which minimise conflicts between traffic and cyclists or pedestrians, avoiding street clutter and where appropriate establishing home zones
- Consider the needs of people with disabilities by all modes of transport

3.8 A key tool is the Travel Plan, as identified in Paragraph 36 of NPPF. All developments which generate significant amounts of movement should be required to provide a Travel Plan.

3.9 Paragraph 37 of the NNPF identifies that planning policies should aim for a balance of land uses that minimise journey lengths for employment, shopping, leisure, education and other activities. Developments should promote a mix of uses in order to provide opportunities to undertake day-to-day activities including work on site.

3.10 When setting local parking standards for residential and non-residential development, Paragraph 39 identifies that local planning authorities should take into account:

- Accessibility of the development
- The type, mix and use of development
- The availability of and opportunities for public transport
- Local car ownership levels
- An overall need to reduce the use of high-emission vehicles

3.11 **DfT Guidance on Transport Assessments:** This document provides advice on the type and scope of a transport assessment and providing guidance on the content of any TA. The advice in this document was considered in the

preparation of the Scoping Note and subsequently in this TA. To ensure a robust assessment specific consideration was given to the following areas:

- Providing an appropriate review of the existing conditions
- Accident analysis
- Provide full details of development proposals
- Provide a 'New Approach to Appraisal (NATA)' assessment
- Assessment years
- Trip Generation
- Promoting smarter travel choices via Travel plans

3.12 **Local Transport White Paper, Creating Growth Cutting Carbon:** This document, dated January 2011, sets out the Government's vision for a sustainable local transport system that supports the economy and reduces carbon emissions. It explains how the Government is placing localism at the heart of the transport agenda, and underlines Central Government's direct support to local authorities, including through the Local Sustainable Transport Fund.

3.13 **Circular 2/07 Planning and the Strategic Road Network:** The circular explains how the HA, on behalf of the Secretary of State for Transport, will participate in all stages of the planning process with Government Offices, regional and local planning authorities, local highway/transport authorities, public transport providers and developers to ensure national and regional aims and objectives can be aligned and met.

3.14 The Circular identifies that the HA will work with developers to secure delivery of proposals in such a way that they minimise any additional burden on other users of the strategic road network. The HA is committed to the promotion of sustainable development and proposals that minimise impact on the Strategic Road Network. Paragraph 21 states:

- The HA will offer advice and technical support that will guide the scale and location of proposals in relation to the strategic road network. The Agency will also provide guidance, for incorporation in the plan, on the scale and nature of improvements to the strategic road network and demand management measures that will be considered in order to facilitate development. This guidance will be relevant to both public and private sector investment decisions. In assessing the interaction with the strategic road network, attention should be paid to the advice set out in the Guidance on Transport Assessment.

3.15 **Dft Guidance on Travel Plans:** The DfT guidance on travel plans requires travel demand to be managed, this being achieved largely through a Travel Plan. The main aims of a Travel Plan are to:

- Encourage travel to and from the site by public transport, walking and cycling rather than by car
- Ensure awareness of the availability of alternatives to driving to and from the development by car
- Improve road safety, particularly by reducing traffic speeds, particularly for pedestrians and cyclists
- Improve personal security, particularly for pedestrians and cyclists
- Encourage participation in the implementation, monitoring and evolution of the Travel Plan

3.16 The DfT guidance includes information on examples of travel plans and the following components should form part of any Travel Plan:

- Site design. The layout of the site will be designed to be attractive for walking and cycling.
- Prioritisation and promotion of public transport. Local Bus stops will be upgraded with suitable facilities. Footway routes to and from bus stops will be surfaced, well lit and user friendly.
- Measures to promote and facilitate walking and cycling. Information will be provided on available facilities and

services through notice boards and leaflets.

3.17 **Manual for Streets 1 and 2 (MfS):** The UK Department for Transport (DfT) and the Department for Communities and Local Government (DCLG), with support from the Commission for Architecture and the Built Environment (CABE), commissioned WSP Group, Transport Research Laboratory (TRL), Llewelyn Davies Yeang and Phil Jones Associates to develop Manual for Streets to give guidance to a range of practitioners on effective street design.

3.18 Manual for Streets was published on March 2007. It superseded Design Bulletin 32 – Residential Roads and Footpaths – Layout Considerations (DB32) and the companion guide Places, Streets and Movement, which have now been withdrawn.

3.19 Manual for Streets 1 provides guidance for practitioners involved in the planning, design, provision and approval of new streets, and modifications to existing ones. MfS aims to increase the quality of life through good design which creates more people-oriented streets. The detailed guidance in the document applies mainly to residential streets. However the overall design principles can be applied to all streets within urban areas.

3.20 A street is defined as "a highway with important public realm functions beyond the movement of motor traffic" – i.e. by its function rather than just the road hierarchy.

3.21 Manual for Streets has updated geometric guidelines for low trafficked residential streets, examines the effect of the environment on road user behavior, and draws on practice in other countries. This research undertaken by Transport Research Laboratory (TRL) provides the evidence base upon which the revised geometric guidelines in the Manual for Streets are based, including link widths, forward visibility, visibility splays and junction spacing.

3.22 Manual for Streets 2 - Wider Application of the Principles is the result of collaborative working between the Department for Transport and the transportation industry.

3.23 The aim of the document is to extend the advantages of good design to streets and roads outside residential areas, largely covered in MfS1. By amending the way high streets and non-trunk roads are designed, the fabric of public spaces and the way people behave can be changed. It means embracing a new approach to design and breaking away from inflexible standards and traditional engineering solutions.

3.24 The new guide does not supersede Manual for Streets 1, rather it explains how the principles of the first document can be applied more widely.

3.25 **Design Manual for Roads & Bridges:** The DfT publish a large suite of documents known as the Design Manual for Roads and Bridges, which provide detailed standards and guidance on the provision of highway networks. Published first in 1992, the numerous elements of the suite are periodically updated. The suite provides a comprehensive manual system which accommodates all current standards, advice notes and other published documents relating to the design, assessment and operation of trunk roads including motorways. The standards are routinely adopted by local highway authorities for their local highway network.

3.26 **Swindon Borough Local Plan 2011:** On 19th July 2006 Swindon Borough Council adopted the Local Plan 2011. The adopted plan forms part of the development plan for the area of Swindon Borough and the basis for decisions on land use planning. The Local Plan was adopted under the transitional arrangements of the Planning and Compulsory Purchase Act 2004 for a period of three years ending on 19 July 2009.

3.27 SBC applied to the Secretary of State to save some of the policies within the Local Plan for a further period until such time they are replaced by the Swindon Core Strategy and the Site Allocations Development Plan Documents

3.28 The saved policies relation to Transport include:

- **T1 – Transport requirements of development -** Development shall be permitted where proposals provide access that is appropriate to the scale, type and location, without detriment to highway safety, traffic movement, and the local environment

- **T2 - Provision of bus travel -** Developments shall cater for, and provide the potential to maximise, bus travel within the site and also connections with existing developed areas.

- **T3 – Protection of proposed bus lanes or other bus priority measures -** Development shall be permitted where it would not compromise the implementation of bus lanes or other bus priority measures identified during the plan period, and those defined on the Proposals Map.

- **T4 – Park and Ride -** Sites defined on the Proposals Map shall be safeguarded for park and ride purposes.

- **T5 – Access by foot and cycle -** Development shall be permitted where there is an existing safe and convenient pedestrian and cycle access or provision is made for such access.

- **T6 – Protection by existing and proposed cycle network -** Development shall be permitted where proposals will not harm a planned or existing cycle route or provision of a more satisfactory route is proposed.

- **T7 – National trails -** Development shall be permitted where proposals do not remove, narrow or materially impair the approved line of the Thames Path or Ridgeway National Trails, and/or public rights of access to, or along these National Trails.

- **T8 – Protection of rights of way -** Development that results in the loss of existing rights of way or their disruption shall only be permitted when adequate, acceptable alternative provision or diversions can be arranged.

- **T9 – Protection of rail freight provision -** The railway line adjacent to the Techno Trading Estate and the County Yard Railway Sidings, as defined on the Proposals Map, are safeguarded from development unless it is proven that their future use is shown to be unviable, in the longer term.

- **T12 – Other road scheme -** Land is safeguarded for the routes/improvements set out below and as defined on the Proposals Map:
  a) Croft Road to Hay Lane Link;
  b) Northern Orbital Road, Purton Road to Great Western Way;
  c) Cricklade Road widening and Moonrakers Junction improvement;
  d) Redposts Drive Link.

3.29 **Swindon Borough Council Local Plan 2026:** The Swindon Borough Local Plan 2026 (formerly known as the Swindon Borough Core Strategy) is the main planning policy document for Swindon Borough, providing the planning policy framework to deliver sustainable growth up to the year 2026. The Local Plan identifies where, when, and how development will take place in Swindon.

3.30 Once adopted, the Swindon Borough Local Plan will form part of the Swindon Borough Local Development Framework.

3.31 The Local Plan 'Pre-Submission Draft' and accompanying documents were published for consultation on the 20th December 2012 for nine weeks. Following the public consultation, the Council has submitted the Local Plan, evidence

studies, and all comments received during consultation, to the Secretary of State for Communities & Local Government for an independent examination by a Government Planning Inspector.

3.32 The Inspector will prepare a report that sets out advice to the Council about the soundness and changes, if any, to the Local Plan. The Council will then be able to adopt the Local Plan in accord with the Inspector's advice.

3.33 The Local Plan includes:

- A vision and strategic objectives for the development of Swindon Borough up to 2026
- Clear policies to guide decisions on planning applications
- Broad locations of development and the allocation of strategic sites
- The infrastructure requirements required in support of the proposed Development
- The strategic context for the allocation of specific sites in either a future Council planning document and/or Neighbourhood Plans
- A key diagram, illustrating the main provisions and the strategic site allocations
- An indication of how the plan will be delivered
- Indicators to monitor the progress of the Plan

3.34 The Local Plan provides an overall spatial Vision for Swindon, as indicated below.

> "The Borough of Swindon will become a place where people choose to live, visit and invest. As an important regional centre, Swindon's appeal will stem from having an attractive and well-equipped town that has successfully blended traditional architecture with high quality contemporary buildings that incorporate sustainable design and construction principles.
>
> The achievement of a high quality public realm in the heart of the town linked to the countryside will provide a real focal point for visitors and Swindonians alike.
>
> Swindon will become one of the best business locations in the UK, offering a high quality of life, not just to its residents, but also to those from a much wider catchment area. New jobs will be created and there will be the fullest range of employment opportunities for the whole community.
>
> Swindon will be at the centre of a network of multifunctional green spaces linking the town to the wider countryside. Swindon will have responded to the needs of a growing population in a way that has protected and enhanced our natural and historic environment.
>
> People in Swindon will have the opportunity to live active, healthy and learning lifestyles."

3.35 To achieve the overall vision, 10 strategic objectives have been identified. Those relating to transport are highlighted below:

- **Strategic Objective 1:** High Quality Sustainable Development – to improve the image of Swindon, enable inclusive communities and address climate change by the provision of high quality, well designed and sustainable development.

- **Strategic Objective 2:** Infrastructure - to meet the infrastructure needs for and arising from the growth of Swindon (including health and community needs) in a timely and coordinated manner and being adequately funded.

- **Strategic Objective 7:** Transport - to support Swindon's growth through the provision of a comprehensive and sustainable transport network that is efficient, safe, affordable, accessible and easy to understand, and offers a genuine choice of modes.

- **Strategic Objective 9:** Green Infrastructure - to provide an attractive and inspirational environment to live, work, learn and play, by the provision of a far-reaching network of connected and multi-functional green spaces linked to the wider countryside.

3.36 To support the Strategic Objectives various Policies have been defined.

- **Policy SD1: Sustainable Development Principles -** To enable the delivery of sustainable development and support sustainable communities in the Borough all development proposals will:

> Be of high quality design
> Promote healthy, safe and inclusive communities
> Respect, conserve, and/or enhance the natural, built and historic environments
> Assess and address the impact of climate change through mitigation and / or adaption measures
> Provide or contribute to the assessed local and borough wide infrastructure and service requirements
> Contribute to the retention of jobs and growth of the local economy and complement town centre regeneration
> Be accessible by walking, cycling and/or public transport
> Use land and resources (such as water, energy, minerals and waste) in an efficient and effective way

- **Policy SD2: The Sustainable Development Strategy**
  Recognising its role and function in the wider area development in the Borough will be concentrated at Swindon through a combination of realising development opportunities within Swindon's urban area; and allocated strategic sites at; Wichelstowe, Commonhead, Tadpole Farm, Kingsdown and the proposed New Eastern Villages, Rowborough and expanded South Marston.

- **Policy SD3: Managing Development**
  When considering development proposals, the Council will take a positive approach to reflect the presumption in favour of sustainable development.

3.37 To support the delivery of the Policies, several overall 'Themes' are outlined. Theme 4 relates to Transport. This identifies specific policies to:

- Improve key transport gateways and corridors
- Provide good access to Swindon Town Centre and key destinations
- Reduce severance caused by transport corridors and the dominance of the car on the streetscene
- Minimise the environmental impact from transport (for example, vehicle emissions)
- Minimise congestion and therefore journey time, noise and air quality
- Promote healthy lifestyles and travel choices and maximise opportunities to walk and cycle
- Support good public transport provision
- Encourage innovative transport initiatives for rural areas

- **Policy TR1: Sustainable Transport Networks - The council will use its planning and transport powers to** help reduce the need to travel, and support and encourage the sustainable, safe and efficient movement of people and goods within and through the Borough.

- **Policy TR2: Transport and Development -** New development should be located and designed to reduce the need to travel and to encourage the use of sustainable transport alternatives, particularly walking and cycling, and provide the potential to maximise bus travel. Development shall be permitted where proposals provide access that is appropriate to the scale, type and location without detriment to highways safety and local amenity, and where there is an existing safe and convenient pedestrian and cycle access or provision is made for such access. Development shall be permitted where proposals will not prejudice or impede an existing or planned cycle route, or provision is made for a more satisfactory route. Development that results in the loss of existing rights of way or their disruption shall only be permitted when adequate, acceptable alternative provision or diversions be arranged. Development shall be permitted where proposals do not remove, narrow or materially impair the approved line of the Thames Path or Ridgeway National Trails, and/or public access to them. Developments shall provide appropriate mitigating measures to offset any adverse impacts on the transport network at both the construction and operational stages. Parking provision, including secure cycle and motorcycle parking, should be provided in accordance with the Council's adopted parking standards. To assess and mitigate the impact of development and to promote sustainable travel choices, the following information will be required to support planning applications:

  > A Transport Assessment: where the proposed development is likely to have significant transport and related environmental impact (in accordance with Department for Transport guidance)

  > A Travel Plan: where the proposed development is likely to have significant transport and related environmental impact (in accordance with Department for Transport guidance).

3.38 To deliver the necessary housing and employment growth identified for Swindon, specific development areas have been identified.

- **Policy NC3: New Eastern Villages - including Rowborough and South Marston Village Expansion**
  Land to the East of the A419, as defined on the Proposals Map, is allocated for a mixed-use development. The form of the development shall comprise a series of new distinct villages and an expanded South Marston village. The development shall provide:
  > About 6,000 dwellings at the New Eastern Villages (south of the A420)
  > About 1,500 dwellings at Rowborough (north of the A420)
  > 500 dwellings at South Marston
  > Affordable housing at Rowborough and South of the A420 in accordance with Policy HA2. Separate proposals are made in Policy RA3 in relation to South Marston.
  > High quality public realm including outdoor civic public space
  > Sustainable transport links including:
    - Walking and cycling network improvements that integrate with existing networks and provide good connectivity to the surrounding area;
    - a bus rapid transit link through the district centre that connects the Eastern Villages to Swindon Town Centre as part of phase 1 of the development which includes residential development north of the A420, the District Centre and the employment allocation;
    - additional public transport services to connect with Swindon and internally within the development
    - an improved gateway junction at White Hart to manage additional demand and deliver high quality public realm
    - Improvements to the Oxford Road/Drakes Way and Covingham Road/Dorcan Way transport corridors
    - A green bridge across the A419 near Covingham Drive to provide for walking, cycling and public transport
    - A new road link to the Commonhead Roundabout;

- A new road link under or across the Bristol to London railway line connecting the development north and south, and new and/or improved accesses to the A420 for proposed residential and employment uses;
- A 1,000 (3ha.) space Park and Ride site; and
- Measures to minimise rat-running through existing adjacent villages and east Swindon.

> An extensive green infrastructure network that maximises opportunities for habitat connectivity and enhanced biodiversity including extending the River Cole green infrastructure corridor and connecting with Nightingale Wood

> Sports and leisure facilities, including playing pitches a leisure centre and a 25m swimming pool;

> About 40 hectares of employment land in total (B Class Uses) to be located south and east of the A420 and A419 respectively adjacent to the White Hart Junction, in a form that complements the mixed-use District Centre, and safeguards its connection with the wider residential development in the Eastern Villages and distributed as follows

- 2.5 hectares within B1a use-class to be located at and/or adjacent to the District Centre;
- 7.5 hectares within B1b/c or B2 use-class; and
- 30 hectares within B8 use class.

> about 12,000m2 (gross) of retail floorspace including a high quality District Centre with strong connectivity to the adjacent residential areas, comprising an anchor food store and complementary uses, and a network of Local Centres that offer retail provision of a scale that meets the daily shopping needs of the communities that they serve, including the existing community of South Marston;

> educational requirements comprising:

- a minimum of 8 -forms of entry of secondary provision;
- a minimum of 8 forms of entry of primary provision with early learning facilities;
- a site for a special school for children and young persons with profound, multiple and severe learning difficulties for ages 0-25, and
- measures to manage the demographic peak at primary and secondary schools;

> Community facilities by the means of safeguarded land and/or developer contributions, including where appropriate flexible, multi-purpose buildings for use by the community, the public sector and for worship at the district centre, local centres and the learning campus;

> A health care facility with GP, dentist and pharmacy at the District Centre

> Safeguarded land for a fire station towards the southern part of the site, and

> A sewage treatment works if required.

3.39 Policy RA3 in respect of South Marston is also relevant. It addresses the expansion of the village. It makes provision for respecting the character of the village, for affordable housing, for education, community and recreation facilities at the heart of the village, for green infrastructure. In respect of transport it provides for a new road connection between Thornhill Road and Old Vicarage Lane seeking that the connection should be designed (within the expanded village) so that it does not form a bypass to South Marston. It provides for *"traffic management and sustainable transport measures to minimise the volume of traffic, including traffic passing through the village, and to reduce the impact on the village, in particular at Pound Corner)"*

3.40 **Infrastructure Delivery Plan:** SBC is required to demonstrate that the policies in the Local Plan will be delivered in a sustainable way. To this end, there is increased emphasis on the need to identify infrastructure required to support future growth, resulting from additional housing and employment during the Plan period to 2026. The IDP focuses on those infrastructure items which will require capital expenditure

3.41 The IDP helps to identify:

- Types and costs of infrastructure

- Delivery timetable
- Gaps in funding
- Critical pieces of local and strategic infrastructure.

3.42 In relation to transport infrastructure the **WSP Swindon Transport Strategy (2009)** sets out Swindon's transport aspirations to 2026.

- **Rapid Transit**
  The Northern Rapid Transit link will deliver the first arm of Swindon's Rapid Transit scheme connecting Tadpole Farm and the Northern Development Area with Swindon town centre. The rapid transit routes from Wichelstowe and the Eastern Villages will provide fast and reliable journey times for new communities to Swindon Town Centre and complete the rapid transit network. The rapid transit network will complement the Park and Ride Strategy – additional sites are proposed at Wichelstowe, Eastern Villages and Commonhead.

- **Town Centre**
  The new Town Centre bus exchange and improved bus stops will support economic growth by providing a viable and attractive alternative to the private car for everyday journeys.

- **Eastern Villages including Rowborough and South Marston**
  The Eastern Villages will deliver a new district centre, not just to serve the new community but also the existing residential areas at Covingham, Dorcan and Parks. The main route from the New Eastern Villages into Swindon's Central Area will be via Oxford Road (A420) and White Hart Roundabout. The intention is to deliver an improved junction at White Hart that establishes strong linkages between the Eastern Villages and the rest of Swindon, prioritises access into Swindon Town Centre and improves the public realm. A Green Bridge is proposed that will provide access across the A419 – part of the Strategic Road network and a significant physical barrier – for public transport, walking and cycling. The Commonhead Link (and associated Junction 15 improvements) will provide direct access to the Strategic Road Network (A419 and M4). Junction improvements will also be required on the A420.

3.43 **Draft Supplementary Planning Document 2009 – Eastern Development Area:** The consultation period for the draft Eastern Development Area (EDA) SPD ran for a period of eight weeks from Monday 20 April to Monday 15 June 2009. The SPD has since been redrafted in light of representations made, changes to the emerging Swindon Core Strategy, and to reflect new evidence based studies that were commissioned following the first draft.

3.44 The 2009 SPD identified that the development area could accommodate a potential 12,000 dwellings together with a range of supporting land uses. The purpose of the document is to provide a planning framework setting out the expectations and requirements of development for the EDA.

3.45 The draft SPD identified a set of development principles, as detailed below:

- Deliver a diverse living environment of high quality that accommodates growth targets within an efficient and generally compact urban form with a strong sense of place.
- Create improved eastern approach routes into Swindon and help redefine the image of Swindon.
- Form part of Swindon and integrate as seamlessly as possible with the eastern edge of town.
- Create new sustainable communities and facilities that connect with and protect the character of existing surrounding communities.
- Support the broader Swindon economy with new employment and investment opportunities.
- Become part of and help deliver a new improved strategic transport network for Swindon that encourages public transport, pedestrian and cycle use, seeking to achieve connectivity within and beyond the EDA.
- Assist the physical, social and economic regeneration of East Swindon.

- Conserve and enhance natural systems, watercourses, biodiversity and landscape resources, as part of an integrated green infrastructure proposal.
- Conserve and enhance Swindon's cultural heritage, recognising its key assets and preserving the identity of those assets within the EDA.
- Connect Swindon with its wider landscape setting in a sympathetic way.

3.46 The draft SPD identified three potential broad layouts for the development, as identified below:

- **The Crunch**: The main premise of 'The Crunch' is that the EDA develops as an integral part of Swindon that, as seamlessly as possible, joins new development to east Swindon. 'The Crunch' seeks to overcome severance caused by the A419, which will otherwise be a barrier to integration between the EDA and the existing eastern edge of town.

- **The Stretch**: In this option it is accepted that the major barriers to integration with Swindon's existing urban area are too difficult or expensive to overcome and that the EDA must establish its own distinctive and separate identity. 'The Stretch' effectively turns the EDA into a Separate community on the edge of Swindon. Whilst it may not directly turn its back to the town, it is not driven by a fundamental aim to integrate with the urban area. This means that the physical centre of the EDA is located in the centre of the new development area – stretched away from east Swindon thereby reinforcing its separate identity.

- **The Weave:** This is a hybrid option that shares the basic philosophy of the crunch but recognises that some of the physical severance issues are challenging if they are to be overcome, at least in the short to medium term. 'The Weave' sees the EDA as an integrated part of Swindon but takes a stepped approach to delivering it. One of the important features of 'The Weave' is that it could evolve into 'The Crunch' at some later stage.

3.47 The draft SPD identified that the Crunch was at the time of writing preferred.

3.48 Subsequent to the production of the draft SPD, additional background studies have been carried out. This included SBC commissioning JMP to undertake some initial traffic modelling. This work utilised the A419 Paramics model. The study focused on the potential transport impact of growth to the east of Swindon on the A419, M4 J15 and J16 and on the A420 east of the White Hart Junction. This report considered 7,500 homes and associated services and facilities up to 2026, which corresponds with the timeframe of the Core Strategy. An updated Halcrow SATURN Model has been used as the basis for this PARAMICS assessment.

3.49 This additional study identified that the White Hart junction in its current form would not have sufficient capacity to cater for the anticipated increases in traffic volumes arising from the Eastern Villages even if mitigation is provided elsewhere such as providing a full or partial all vehicular link at the Green Bridge. The report then suggested that the existing roundabout be amended to reflect a diamond interchange.

3.50 A basic comparison of journey times that the operation of a part signalised White Hart roundabout with modified slips would operate just as well / slightly better than the diamond interchange in terms of journey times.

3.51 The report identified that the extent of works required to the A420 is largely dictated by the need to upgrade existing junctions and provide new junctions along the corridor. These junctions generally require two or more approach lanes in each direction. The western site access and Gablecross are situated sufficiently close together as to necessitate the widening of the remainder of the A420 to a two lane standard as a minimum in the westbound direction.

3.52 In total, three existing junctions would need to be upgraded (Gablecross, Police Station and Old Vicarage Lane) and two new junctions constructed in order to provide a western and eastern access into the development.

3.53 The existing junctions would need to be upgraded so as to provide a balance between demand for A420 through movements and that from the Eastern Villages and South Marston. At Gablecross the replacement of the existing roundabout with a signalised crossroads would provide additional capacity and would facilitate pedestrian movements.

3.54 The report identified that the Commonhead to M4 J15 improvements associated with the Commonhead planning application should be sufficient to accommodate the Eastern Villages if the focus of the development is along the A420.

3.55 **Eastern Villages Draft Supplementary Planning Document 2013 :** The draft SPD replaced the 2009 Draft SPD for the Eastern Development Area. It identified a set of development principles, as detailed below:

- Sets out the overall vision and principles
- Seeks to ensure that the Vision for the development is delivered in accordance with the key principles
- Provides a detailed planning framework
- Provides a catalyst to express aspirations for the development
- Offers detail and clarity on the expected outcomes from the development

3.56 The SPD identifies the overall Vision for the Eastern Village Development. The Eastern Villages development will create sustainable well-designed places where people want to live, work and spend time. The development will:

- Be a high quality, sustainable development with strategic infrastructure that benefits Swindon as a whole
- Sensitively and positively respond to the existing landscape context, natural and historic assets and the character and identity of the surrounding villages as well as enhancing biodiversity and green infrastructure;
- comprise new distinct villages with individual identities and characters linked together by green spaces that integrate with the existing urban area and wider landscape setting;
- provide the facilities and services required for the new communities and opportunities for existing communities to benefit from the development
- form a new eastern gateway to Swindon that improves the image of the Town and maximises benefits to the wider economy including connections to the Town Centre.

3.57 The key principles will be to:

- Create vibrant, attractive places that are functional, durable and capable of adapting to accommodate changing lifestyles in line with sustainable communities
- Ensure the timely delivery of necessary facilities to support these sustainable communities
- Conserve and enhance natural systems, watercourses, biodiversity and landscape settings, including integration with green infrastructure to help mitigate and facilitate adaptation to climate change
- Conserve and enhance Swindon's historic assets and cultural heritage, allowing their use and enjoyment without harming them for future generations
- Ensure the development becomes part of the wider town and supports the physical, social and economic regeneration of East Swindon. Respect and support the existing character and setting of surrounding villages;
- Contribute to and not harm the Town Centre regeneration and support the broader Swindon economy with new employment and investment opportunities;
- Contribute to an improved strategic transport network for Swindon and facilitate public transport priority for journeys into the Town Centre;
- Encourage walking and cycling, within the development and ensure high levels of accessibility and connectivity within and beyond the new Eastern Villages
- Enhance the image of Swindon by maximising place-making opportunities particularly around key nodes, gateways and frontages.

3.58 The SPD identifies that the development will deliver approximately 1,500 dwellings will be located at Rowborough, 500 new homes at an extension to South Marston and approximately 6,000 dwellings south of the A420.

3.59 In relation to Transport the SPD identifies the interventions identified with policy NC3 of the Local Plan, viz:

- Improved White Hart junction
- Improvements to the existing network including A420
- A green bridge
- Park and Ride site
- Rapid transit link and additional public transport services
- Measures to minimise rat running through existing communities
- New road link to Commonhead

3.60 The SPD sets out a preferred long term solution for the Improvement of the White Hart roundabout which is described as the diamond interchange. Further detailed design and the securing of funding is required. The SPD recognises nevertheless that the Eastern Villages may proceed by means of the "phased implementation of improvements".

3.61 With regard to the A420, the improvements required are, unsurprisingly, consistent with those set out in the Local Plan and South Marston SPD (see below) and include:

- a new eastern access to Rowborough (north);
- an eastern access to the Park and Ride site in the south
- improvements to the junction at Old Vicarage Lane;
- a new access to serve the new employment area (to the south);
- improvements to the Police Station junction and improvements to Gable Cross junction.

3.62 **South Marston Draft Supplementary Planning Document(June 2003) and Background Transport Assessments** The purpose of the SPD is to provide a planning framework setting out the expectations and requirements for the sustainable development of the village of South Marston. In preparation of the SPD, SBC commissioned JMP to investigate the existing challenges and to review the potential impact of the Eastern Villages.

3.63 The JMP initially identified a range of potential issues with in the transport system together with possible interventions, as detailed below.

Description	Constraint	Intervention
Pedestrian and vehicular conflict around Pound Corner	Insufficient carriageway and footway available	Realignment of the carriageway at Pound Corner to provide footway on the northern side of Old Vicarage Lane
Gaps in footway provision	Thornhill Road close to the industrial estate. Vicinity of South Marston Hotel	Continuous footway from Pound Corner to Keypoint roundabout and from Pound Corner to South Marston Hotel
Accessibility improvements	Lack of Dropped Kerbs / Tactile paving	To provide dropped kerbs / tactile paving in locations where these are not provided
High Traffic Speeds/ Through Traffic	Little evidence to support view in terms of speeds/accidents or traffic volumes.	20mph zone for village centre & 30 mph between village centre and A420 for consistency. Some additional traffic calming measures

Figure 3a: Existing constraints

3.64 The JMP study then investigated the potential impact of the delivery of the Eastern Village concentrating on the impact on the A420 and within South Marston. This report notes that increase in traffic arising from development are from a very low base. The report concluded that "with the exception of the junctions along the A420, junctions within South Marston are expected to operate with minimal queuing. This assumes that a package of measures is in place for the A420 and A419 which deters vehicles from rat running through the village" (JMP Technical Note: *"South Marston Supplementary Planning Document Part Three Transport Assessment",* June 2012". Rat running is therefore avoided, on JMPs analysis, providing journey times along the A419 and A420 continue to make these routes attractive to use. The junctions with the A420 are identified as needing interventions to ensure that this is the case.

3.65 A key conclusion of the JMP technical note, reinforcing the conclusion regarding transport impacts in South Marston itself was that a roundabout junction on Old Vicarage Lane – providing access from Rowborough would be "unlikely to result in a significant increase in traffic volumes through South Marston village"(para 5.6) since most traffic is likely to travel south.

3.66 The study also identified options for improvements at Pound corner involving one or two way traffic and seeking to widen the existing footways noting that there were pros and cons of each alternative.

3.67 The purpose of the South Marston Supplementary Planning Document (SPD) is to provide a detailed planning framework for South Marston, setting out the expectations and requirements of development within the village. It addresses the existing and the expanded village. The SPD describes a vision that: "*South Marston should be a high quality, integrated village with an enhanced sense of community, activity and safety within a distinctive form that builds on the existing character and features in a sustainable manner*". In order to deliver the vision a number of issues will need to be addressed including, in relation to transport: "*improvement of existing, and provision of new highway infrastructure, including resolving issues of through-traffic, Pound Corner, the absence of road-side paths, access onto the A420 and off-site provisions to ensure free movement along the A420*"(para 3.3).

3.68 The Development Principles in respect of Access and Movement are set out in para 6.28 to 6.46. The SPD:

- Requires a Transport Assessment (TA), to be undertaken by the developer(s), to determine the number of additional trips generated and how these will be accommodated on the existing highway network; and
- Expects a sustainable Travel Plan to accompany the TA to promote alternatives to single occupancy car use.

3.69 The SPD promotes:

- A 20mph speed restriction throughout the village (30mph outer limit) with psychological traffic calming and new gateway features.
- The retention of Rowborough Lane and Nightingale Lane as public rights of way/bridleways, with no new development being served by either.
- Junction improvements on A420 (including at Gablecross and at the Police Station access, with the introduction of a new eastern access near Acorn Bridge and capacity improvements along the A420
- Improvements to Pound Corner including for instance the provision of a raised footway on the northern side of Pound Corner;
- An all movement junction to link Rowbrough to Old Vicarage Lane is proposed consistent with the advice of JMP;
- The re-alignment and signalisation of the junction of Old Vicarage Lane and the A420 in order to provide a safe egress onto the A420, particularly for those vehicles travelling west towards White Hart Junction.
- The design of highways as a series of streets with important public realm functions, rather than as roads whose primary purpose is the movement of traffic.
- Enhanced footpath and cycle connections.

3.70 The highway principles and technical work determine that there is no need of a n eastern bypass for South Marston and that connections between Old Vicarage Lane and Thornhill Road should not be considered a "southern bypass" for the village

3.71 **SBC Local Transport Plan 3 (LTP3):** The Local Transport Plan 3 was adopted in March 2011 and covers the period from 2011- 2026. It sets out an overarching strategy for the delivery of the Council's transport programme. The mission statement for the LTP3 is:

> *"a safe, effective and fit for purpose transport network that supports Swindon's ambitions for town centre regeneration and economic growth whilst protecting and enhancing quality of life and the environment for the benefit of local residents, visitors and businesses"*

3.72 While LTP3 covers maintenance and small schemes, which are not identified within the IDP, larger more strategic transport schemes are included.

4 Existing Transport Conditions

Existing Travel Behaviour Overview

4.1 A review of 2011 Census data has identified the existing mode share for residents within all of the existing Wards. The modal split is indicated in Figure 4a with the distance travelled to work is indicated in Figure 4b. The site is largely located in the Blunsdon Ward and neighbouring the Wards of St Margaret's, St Phillip and Covingham & Nythe.

4.2 Blunsdon Ward contains a large geographically area but only includes 1,323 dwellings accordingly to 2011 Census data. As a result of the limited number of dwellings, the Blunsdon ward is not deemed representative of the proposed development characteristics. The neighbouring St Margaret's Ward is located in a similar position to the highway network and would access similar sustainable facilities. In regard to the character of the St Margaret's ward, there are 4,321 households of which 3,034 are identified as family households. The household type is primarily 3 bed units supported by 2 and 4 bed units. It is considered that the St Margaret's ward is representative of the proposed development.

4.3 Based on the 2011 data, the most dominant mode of travel is by car, resulting in 69.5% of all trips within the St Margaret's ward. Travel by foot is the second most dominant mode of travel.

4.4 The 2001 Census data also provides an indication of the distance travelled to work. This indicates that 7.6% of working people work home. This percentage is largely representative of the majority of wards in the area and is consistent with the average for all of Swindon. 75.3% of the work force travel less than 10km. Therefore a large proportion of the workforce remain within the Swindon hinterland.

Mode	Percentage
Train	0.9%
Bus, minibus or coach	5.7%
Driving a car or van	69.5%
Passenger in a car or van	5.6%
Motorcycle, scooter or moped	1.3%
Taxi	0.4%
Bicycle	5.7%
On foot	7.0%
Work mainly at or from home	3.4%
Other method of travel to work	0.5%

Figure 4a: Mode Share

Mode	Percentage
Working at or from home	7.6%
Less than 2km	27.0%
2km to less than 5km	40.7%
5km to less than 10km	10.1%
10km to less than 20 km	2.2%
20km to less than 40km	3.1%
40km to less than 60km	2.4%
60km and over	3.3%
Other	3.5%

Figure 4b: Distance travelled to work

Existing Highway Network

4.5 The location of the site in relation to the local road network is indicated in Figure 4c.

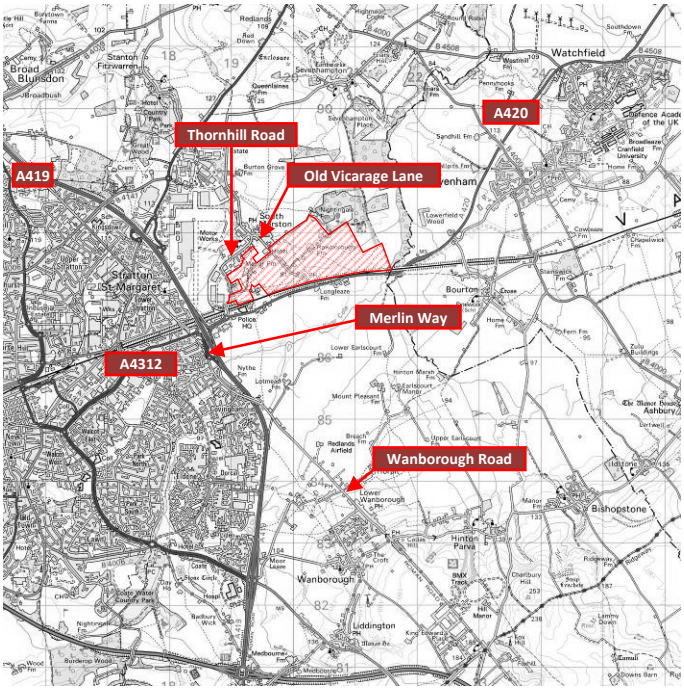

Figure 4c: Site Location in relation to the local road network

4.6 The majority of the road network adjacent to the site is classified as part of the Local Road Network (LRN), managed by SBC, with the A419 and the M4 Motorway managed by the HA. The roads managed by the HA are typically identified as being a part of the Strategic Road Network (SRN).

4.7 The A420 is located to the south of the site and forms an important corridor within the LRN. The A420 commences from Swindon and heads generally in a north easterly direction extending into Oxford. The A420 is predominantly single carriageway road which is subject to 50mph speed limit along much of the length. There are sections of dual carriageway and the section approaching Swindon is subject to a 50mph speed limit.

4.8 There are a several junctions along the length of the A420 which are predominantly at-grade. The junctions along the A420 that are within the hinterland to the site are:

- A420 / Old Vicarage Lane
- A420 / Police HQ
- A420 / Gable Cross
- A420 / A419

4.9 Old Vicarage Lane forms a priority T-junction with the A420 with a ghost island right turn bay for vehicles travelling along the A420. From the junction with the A420, Old Vicarage Lane continues in a north westerly direction towards the village of South Marston. Old Vicarage Lane is rural in character and does not provide any dedicated facilities for pedestrian or cyclists. Circa 100m north of the A420, Old Vicarage Lane passes under the Swindon to Paddington rail line, as indicated below. The carriageway beneath the underbridge is of reduced cross sectional width.

Figure 4d: Southerly view along Old Vicarage Lane

4.10 To the north, Old Vicarage Lane forms simple priority junctions with Nightingale Lane and Rowborough Lane. These are minor roads which provide access to a small number of existing residential properties.

4.11 Circa 1.4km to the west of the junction with Old Vicarage Lane, the A420 provides a signalised access junction to the Wiltshire Police headquarters. There is no facility to undertake a right turn from the A420 into the police headquarters, so vehicles needing to access the facility from an easterly direction must continue through this junction toward the Gablecross roundabout and undertake a U-turn. Heading in a westerly direction from this junction, there is an off road segregated walking and cycling route.

4.12 The Gablecross roundabout lies circa 200m to the west of the Police HQ signals and is a five arm roundabout. The arms of the roundabout serve Sainsbury's supermarket to the south, Thornhill Road and the St Margaret's Retail Park to the north with the remaining arm being exit only from the retail park. The off-road pedestrian/cycle route continues westerly through Gablecross junction. There is an unsignalised pedestrian crossing across the Thornhill Road arm with a signalised pedestrian crossing to the west of the junction to provide a safe route into the supermarket.

4.13 Thornhill Road heads in a generally northerly direction to access the South Marston village. Thornhill Road is rural in nature and does not provide for dedicated facilities for pedestrian or cyclists within the highway over much of the length, as indicated below. There is however a pedestrian and cycle route alongside Thornhill Road between the Gablecross roundabout and the Keypoint roundabout and there is a segregated pedestrian and cycle route to the west of Thornhill Road between the Keypoint roundabout and the start of south Marston village. There is traffic calming (single file traffic) at the entry and exit of the village which attempts to control vehicle speeds into South Marston

4.14 Thornhill Road continues into the village of South Marston where there are sections of a discontinuous footway. In the centre of South Marston, Thornhill Road forms a priority T junction with Old Vicarage Lane, with this junction known locally as Pound Corner. The village is accessed by three roads (Highworth Road, Thornhill Road and Old Vicarage Lane) that link to the north with the A361 and to the south with the A420 and onward to the A419 trunk road and M4. These roads are considered to be potential "rat runs" between the strategic roads and to the major employment areas adjacent to the village. The alignment and relative narrowness of these roads create points of potential pedestrian and vehicular conflict, including Pound Corner.

Figure 4e: Northerly view along Thornhill Road

4.15 Circa 300m to the west of the Gablecross junction lies a grade separated junction between the A419 and the A420, known as the White Hart interchange. The interchange has two parts that are connected by part of Merlin Way. The northern part of the interchange is a grade separated roundabout with the A419 main carriageway passing over the roundabout. The roundabout connects the A420, the A419 southbound slip roads on the east side of the roundabout together with Oxford Road and Ermine Street to the west and Merlin Way to the south. The northbound A419 on and off slips have a roundabout junction with Merlin Way that is located 400m to the south of the White Hart roundabout junction and form the southern part of the interchange . Any vehicles going to the north on the A419 from the A420 have to travel southwards along Merlin Way before travelling north along the A419 and as such the present arrangement is counter intuitive. Oxford Road continues from the White Hart interchange westerly towards Swindon town centre.

4.16 The White Hart interchange incorporates a pedestrians and cyclist bridge linking the A420 movements east-west either side of the A419.

4.17 A Toys-R-Us store is located to the south east of the A419. Inbound access to the store is taken direct from the A419 southbound on slip. Outbound trips joining the A420.

4.18 Merlin Way continues south from the White Hart interchange and becomes Wanborough Road which crosses over the A419 via a single carriageway road bridge 1.5km south of White Hart Roundabout. Wanborough Road links with Wanborough and Upper Wanborough, and has one lane per direction.

4.19 Continuing from the White Hart interchange, Oxford Road (A4312) provides a westerly link into Swindon town centre. Oxford Road is a single carriageway road with a footway to the north and a footway/cycle way to the south. Oxford Road is a residential road allowing direct access to individual properties. Oxford Road has a signalised pedestrian crossing near Wanborough Road, a signalised junction with Nythe Road and a roundabout junction with Dorcan Way, with this junction being identified as the Green Bridge roundabout, being adjacent to the Green Bridge Retail Park.

4.20 From the White Hart interchange the A419 heads in north westerly and southerly directions. To the north west the road heads toward Cirencester passing through further junctions identified locally as the Honda and Rat Trap roundabouts. To the south, the road passes through the Commonhead junction to link with the M4 at Junction 14. The A419 takes the

form of a lit dual-carriageway which is subject to a national speed limit of 70mph. There are no pedestrian or cyclist facilities along the A419.

4.21 The M4 motorway caters for east – west movements and provides the major ink between South West England / Wales to the west and Reading / London to the east. The grade separated junction with the M4 is signalised. The roundabout is lit and there are no pedestrian or cyclist facilities at the junction.

Existing Sustainable Facilities and Services

Pedestrians and Cyclists

4.22 At the present day, the development land does not have any significant traffic generators within the boundary. As such, there has been no material requirement for footway / cycleway provision on site or to access the site. The nearby primary roads that do provide footways have been identified earlier in this section.

4.23 Swindon has a cycling network consisting of a mixture of both on and off road routes that also caters for walking trips. As indicated earlier, an off road cycling route commences from the police headquarters and continues westerly towards the town centre. A cycle way extends northwards from the junction with Gablecross to link South Marston. At the A420, the White Hart Interchange provides a bridge for onward travel towards the town centre. There are additional routes to the east of the site.

4.24 The figure below identifies the location of the cycleways closest to the site.

4.25 The desire line from the site is predicted to be westerly towards the supermarket and the town centre, where existing provision is available.

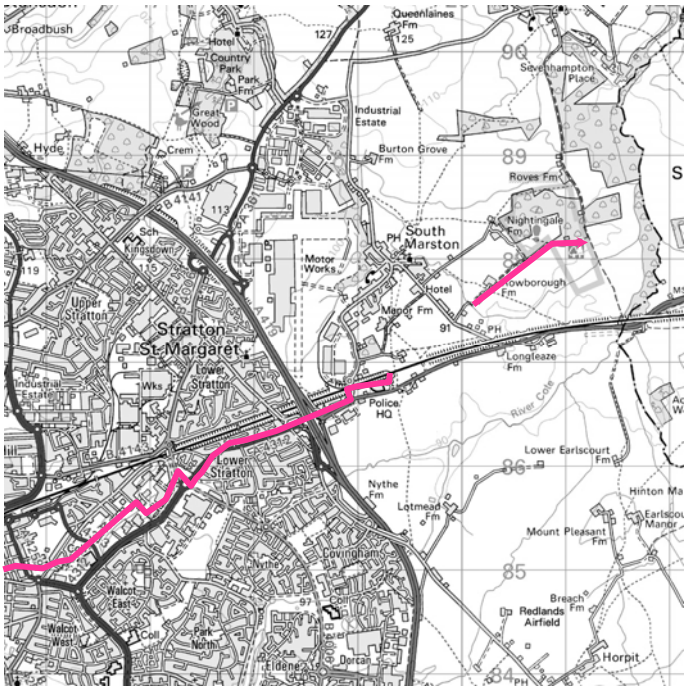

Figure 4f: Cycleways closest to site

Public Rights of Way

4.26 Public Rights of Way (PRoW) are classified as highways and as such are protected routes. The 1949 National Parks and

Access to the Countryside Act placed a duty on every County Council in England and Wales to draw up and publish a definitive map and statement of PRoW in their area.

4.27 The Definitive Map is the legal record of the location and status of PRoW. The statement is a description of the PRoW shown on the definitive map.

4.28 There are four classifications of PRoW:

- Footpaths - by foot only
- Bridleways - by foot, horse or bike
- Restricted byways - by any form of transport that doesn't have a motor
- Byways open to all traffic - let you travel by any form of transport, including cars

4.29 The figure below highlights the PRoW that are closest to the site. This illustrates that there is a network of bridleways and footpaths that cross the site and connect with bridleways that penetrate other roads in the vicinity of the site. In particular there are existing strong links between South Marston and the A420 for pedestrians and cyclists to Gablecross and for pedestrians across the railway at underbridges further to the east near the Police HQ and to the west and east of Old Vicarage Lane.

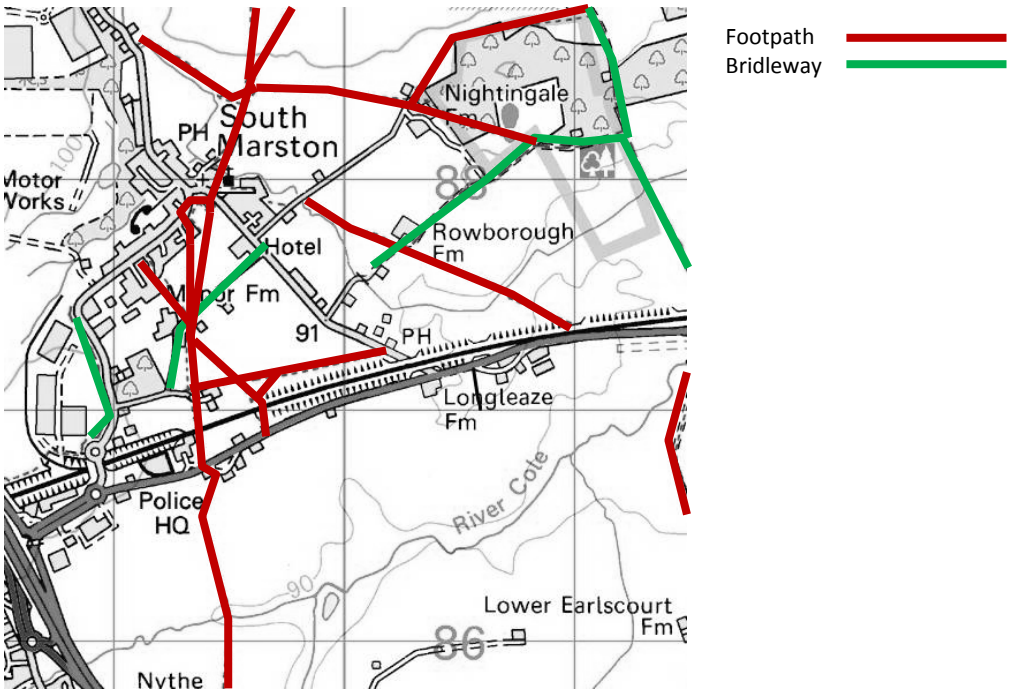

Figure 4g: Public Right of Way

Public Transport - Road

4.30 Bus services 65 and 66 operate along the routes close to the site as indicated below.

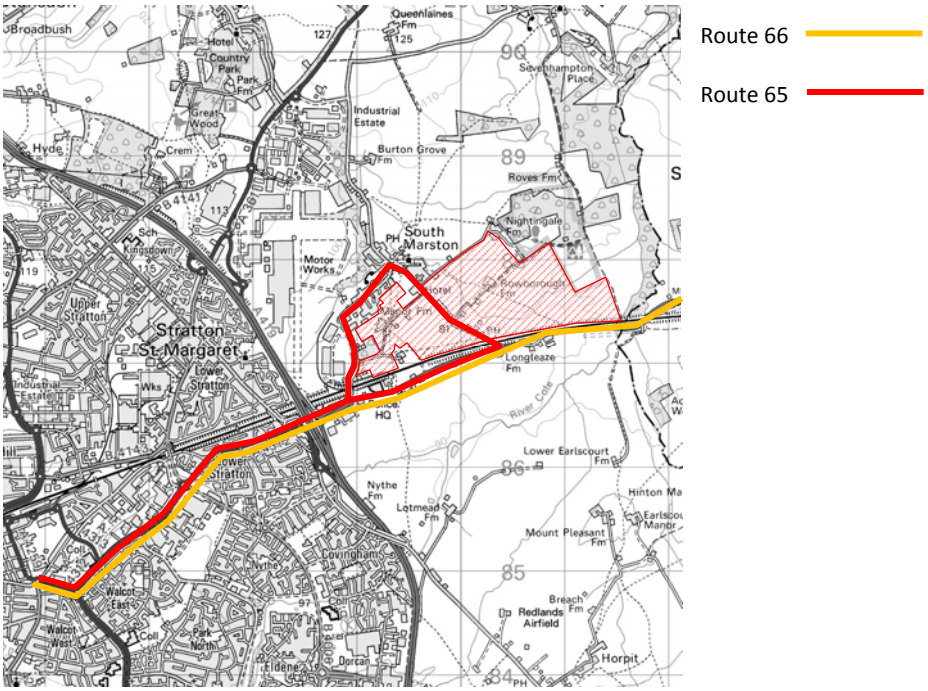

Figure 4h: Bus Routes operating close to the EV

4.31 The bus service 65 is a route managed by Stagecoach and operates between Swindon and Faringdon, Monday to Saturday. The first bus leaves Swindon Bus Station at 10:50 calling at South Marston at 11:03. For trips towards Swindon, the first bus calls at South Marston at 10:26 with a journey time of 15 minutes. This service operates typically four times daily.

4.32 The Route 66 is managed by Stagecoach and operates along the A420 between Swindon and Faringdon, Monday to Sunday. The first bus leaves Swindon at 05:40 and typically operates on a half hourly frequency. The journey time to Oxford is 1 hour and 15 minutes.

4.33 The bus station provides waiting areas, toilets and is located circa 750 metres to the south of Swindon train station.

Public Transport - Rail

4.34 The closest train station is located in central Swindon, circa 6km to the west of the proposed site. The train station, presently managed by the franchise operator, First Great Western, provides a range of facilities including:

- 578 space car park open 24 hours
- Ticket office open 7 days of the week
- Self-service ticket office
- Manned help desk
- Cash machine
- Pay phones
- Post box
- Public WiFi
- Refreshments
- Shops
- Toilets
- Waiting rooms

4.35 Swindon rail station provides three routes per hour to Didcot with a journey time of circa 17 minutes. From Didcot, onward travel to Oxford is possible. Additional routes to London Paddington are provided three times per hour with a journey time of 63 minutes.

4.36 Trips from Swindon westwards to Bristol are supported, with two trips per hour each to Temple Meads (45 minute journey time) and to Bristol Parkway (Journey time 27 minutes).

Accident Review

4.37 Data has been obtained from Wiltshire Police for all recorded personal injury road accidents (PIAs) occurring during a five year period up to the end of February 2013 on the A419, A420, A4312 and A361 in the vicinity of the site.

4.38 A total of 420 accidents are recorded, which are summarised below, resulting in 600 casualties. Two accidents resulted in fatal injury, both to a motorcyclist. A further 51 accidents resulted in serious injury.

	Number of PIAs				Casualties
	Slight	Serious	Fatal	Total	
Year 1 to end Feb'09	88	13	0	101	136
Year 2 to end Feb'10	75	14	0	89	121
Year 3 to end Feb'11	62	6	1	69	99
Year 4 to end Feb'12	71	11	0	82	119
Year 5 to end Feb'13	71	1	1	79	125
5 year period total	367	51	2	420	600

Figure 4i: Total number of PIAs by year and severity, with casualties

4.39 A notable statistical outlier relates to the high proportion of rear end shunts, making up 44% of all accidents. Also, 20% of the accidents involved vehicles skidding.

4.40 Other notable factors are that 131 of the accidents (representing 31%) involved at least one young driver or rider aged 23 years or under and that 30% of the accidents involved vulnerable road users i.e. pedestrians and the riders of 2-wheeled vehicles. The numbers of such accidents are listed below.

	Number of PIAs				
	Under 17	17-18	19-20	21-23	TOTAL
Pedal cyclist	2	4	2	2	10
Motor cyclist	6	3	5	5	19
Car/LGV driver	0	25	33	41	99
HGV 7.5t	0	0	0	1	1
5 year period total	8	32	40	49	129

Figure 4j: PIAs involving young drivers/riders

	Number of PIAs			
	Pedestrian	Pedal cyclist	Motor Cyclist	Total
Year 1 to end Feb'09	2	10	13	25
Year 2 to end Feb'10	4	10	17	31
Year 3 to end Feb'11	2	5	6	13
Year 4 to end Feb'12	9	8	12	29
Year 5 to end Feb'13	6	7	9	22
5 year period total	23	40	57	120

Figure 4k: PIAs involving vulnerable road users

4.41 For the purpose of more detailed analysis, the accidents have been split by area and shown in the table below. Significant clusters within each area have then been identified and studied in greater detail. These have been compared against expected accident rates from COBA for that particular type of road layout and volume of traffic.

	Slight	Serious	Fatal	Total
Area 1: M4 Junction 15	32	4	0	36
Area 2: A419	75	8	1	84
Area 3: A420 E of A419	43	4	0	47
Area 4: A4312 W of A419	64	4	0	68
Area 5: A361 E of A419	27	0	0	27
Area 6: Other - main roads	78	20	1	99
Area 7: Other - unclassified	48	11	0	59
5 year period total	367	51	2	420

Figure 4l: PIAs involving vulnerable road users

Area 1/Cluster 1 – M4 Junction 15 Interchange with A419 and A346

4.42 This area of the road network covers the whole of the M4 Intersection with the A419 and A346, including the immediate junction approaches and is shown as "Cluster 1" on Figure 1.

4.43 A total of 36 PIAs are recorded at this location during the 5-year study period, equivalent to 7.2 PIAs per annum. Of these, none resulted in fatal injury but four resulted in serious injury, three to motorcyclists. These serious accidents do, however, appear to be on a falling trend with two occurring during the first 12 month period and one in each of the two subsequent years and none in either of the most recent two years.

4.44 There were a total of 52 casualties, averaging 1.4 per PIA.

4.45 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 9.0 PIAs per annum which compares with an observed rate of 7.2. This suggests that the junction is performing better than might be expected and that overall the results reported give no particular cause for concern.

4.46 The only notable feature of the accidents is that 25 or 67% of the accidents involved rear end shunts. Thirteen of these occurring on the A419 southbound approach and entry to the roundabout.

Area 2 – A419 between M4 Junction 15 and Blunsdon

4.47 This section of the road network covers the whole of the A419 between the M4 motorway to the south and Blunsdon to the north.

4.48 A total of 84 PIAs were reported at this location during the 5-year study period, equivalent to 16.8 PIAs per annum. Of these one resulted in fatal injury and eight resulted in serious injury. The accidents do, however, appear to be on a falling trend with 27, 20, 19, 11 and 7 occurring in each of the 12-month periods studied. The eight serious injury accidents are also on a falling trend with three, two, one, one and one in each 12-month period.

4.49 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.174 PIA/mvkm which compares with an observed rate of 0.071. The overall accident record, therefore, would appear to relatively good.

4.50 Notable factors amongst the 84 reported accidents are that 64% of them involved rear end shunts and 38% involved at least one vehicle skidding. 15% involved HGVs.

4.51 One fatal accident occurred in April 2010 just north of Commonhead Roundabout when a pedal cycle travelling south with no lights was hit by a heavy goods vehicle (HGV), displacing the rider who was then run over by the HGV and three following vehicles. The fact that the rider was wearing dark clothing and displaying no lights is mentioned as a contributory factor in the accident although lighting conditions were not given as being "dark".

4.52 Two clusters of accidents have been identified within this area worthy of further analysis, i.e. the section of the A419 between the M4 motorway and White Hart Interchange with the A420 and White Hart Interchange itself. These are shown respectively as "Cluster 2" and "Cluster 3" in Appendix B. Cluster 2 includes the fatal accident.

4.53 **Cluster 2 – A419 between M4 and White Hart Interchange:** On this 4.0km section of the A419 24 PIAs were reported including the fatal accident described above. None of the other accidents resulted in either fatal or serious injury.

4.54 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.174 PIA/mvkm which compares with an observed rate of 0.059. The overall accident record, therefore, would appear to relatively good.

4.55 Notable factors amongst the 24 reported accidents are that 58% of them involved rear end shunts 42% occurred after dark. 50% involved at least one vehicle skidding and 38% occurred on a wet road surface. There are no further notable clusters within this section.

4.56 **Cluster 3 –White Hart Interchange:** At this Intersection 44 PIAs were reported including two that resulted in serious injury.

4.57 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 5.7 PIAs per annum which compares with an observed rate of 8.8. This suggests that the junction is performing worse than might be expected in terms of safety.

4.58 By far the most notable pattern amongst these accidents is that 89% of them involved rear end shunts, mainly at the three major entries to the roundabout:

- From A419 southbound off-slip – 15 PIAs, all involving rear end shunts,
- From A420 westbound – 17 PIAs, all but one involving rear end shunts,
- From A420 Merlin Way – 9 PIAs, all but one involving rear end shunts.

4.59 The two principal causation factors given for these accidents are, “failed to look properly” and “failed to judge other persons path or speed”. There is clearly an issue at this junction concerning drivers not slowing sufficiently on the approaches and/or judging correctly when it is safe to proceed.

4.60 Proposals outlined elsewhere in this report to signalise two of the three entries onto White Hart Interchange, where this problem currently exists, is likely to have safety benefits in that with a more formalised junction control drivers’ decisions are made easier and there is less room for misjudgement to occur.

Area 3 – A420 east of White Hart Roundabout

4.61 This section of the road network covers 3.0km of the A420 between the A419 White Hart Interchange and the county boundary to the east.

4.62 A total of 20 PIAs were reported during the 5-year study period, equivalent to 4.0 PIAs per annum. Of these none resulted in fatal injury but two resulted in serious injury.

4.63 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.381 PIA/mvkm which compares with an observed rate of 0.168. The overall accident record, therefore, would appear to relatively good.

4.64 Notable factors amongst the 20 accidents are that 35% of them involved rear end shunts and 35% occurred on a wet road surface. Seven of the accidents involved vulnerable road users: one pedestrian, four pedal cycles and two motorcycles.

4.65 Three small clusters have been identified on this section of road. They are small but in view of their proximity to the development site they have been analyses separately in order to identify any addressable issues. These are shown on Figure in Appendix B as Clusters 4, 5 and 6.

4.66 **Cluster 4 –A420 Gable Cross Roundabout:** Four accidents were reported at Gable Cross Roundabout, one of which resulted in serious injury to a pedal cyclist when a car entered the junction from Thornhill Road, failing to give way to the pedal cycle, which was already on the roundabout. Another of the accidents involved a pedal cyclist travelling west who failed to give way to a car already on the roundabout. Another accident involved a motorcyclist travelling east who hit a car whilst changing lane on the exit from the roundabout. The fourth one involved and was caused by an unmarked police car also changing lane on the eastbound exit.

4.67 Although three of the four accidents involved vulnerable road users there are no clear common factors to suggest any particular problem.

4.68 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 1.8 PIAs per annum which compares with an observed rate of 0.8 which is considerable lower than this.

4.69 **Cluster 5 –A420 in vicinity of Esso Service Station:** Five slight injury accidents were reported in the vicinity of the Esso service station between Gable Cross Roundabout and the priority junction with Vicarage Lane. One of the accidents involved a single vehicle losing control on an icy road surface. Three of the others involved a rear end shunt in slow moving traffic, two eastbound and one westbound. The fifth accident involved a vehicle overtaking a slow moving vehicle and hitting a tractor-with-trailer turning right ahead of it.

4.70 None of the accidents appear to have resulted directly from activity associated with the filling station but all are a by product of traffic slowing down in the general vicinity.

4.71 **Cluster 6 –A420 junction with Vicarage Lane:** Three slight injury accidents were reported at the junction of Old Vicarage Lane with the A420. One of the accidents involved a rear end shunt on the eastbound approach to the junction. The other two both involved single vehicles, one losing control turning left into Old Vicarage Lane on an icy road surface and the other whilst swerving to avoid an animal in the road. These latter two accidents resulted from unusual conditions and there does not appear to be a problem specifically related to this junction.

4.72 The number of accidents per annum expected at a junction of this type and traffic level based upon COBA parameters would be 0.8 PIAs per annum, which compares with an observed rate of 0.6, indicating that the junction is performing within acceptable parameters.

Area 4 – A4312 and A4259 west of A420

4.73 This 3.5km section of the road network covers the A4312 between the A419 in the east and Drakes Way in the west, and the A4259 just south of the A4312.

4.74 A total of 68 PIAs were reported at this location during the 5-year study period, equivalent to 13.6 PIAs per annum. Of these none resulted in fatal injury but four resulted in serious injury. These serious accidents do, however, appear to be on a falling trend with one, three, zero, zero and zero in each of the 12-month periods.

4.75 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.884 PIA/mvkm which compares with an observed rate of 0.523. The overall accident record, therefore, would appear to relatively good.

4.76 Notable factors amongst the 68 reported accidents are that 49% of them involved rear end shunts and 35% involved drivers/riders aged 23 years of age or under. 29% of the accidents involved vulnerable road users: two pedestrians, six pedal cycles and 12 motor cycles.

4.77 There are no notable clusters of accidents along the route.

Area 5 – A361 east of the A419

4.78 This section of the road network covers the A361 between the A419 and 3.0km to the east.

4.79 A total of 27 PIAs were reported at this location during the 5-year study period, equivalent to 5.4 PIAs per annum. Of these none resulted in either fatal or serious injury.

4.80 The number of accidents per annum expected along a road of this type and traffic level based upon COBA parameters would be 0.381 PIA/mvkm which compares with an observed rate of 0.250. The overall accident record, therefore, would appear to relatively good.

4.81 The only notable factor amongst the 27 reported accidents are that 37% of them involved vulnerable road users: zero pedestrians, four pedal cycles and six motor cycles.

4.82 There are no notable clusters of accidents along the route.

Area 6 – Other main roads

4.83 This area covers all main roads of the study area that are not included in the sections above.

4.84 A total of 99 PIAs were reported at this location during the 5-year study period, equivalent to 19.8 PIAs per annum. One accident resulted in fatal injury and 20 in serious injury.

4.85 Notable factors amongst the 99 reported accidents are that 40% of them involved vulnerable road users: nine pedestrians, 14 pedal cycles and 17 motorcycles.

4.86 One fatal accident occurred in July 2012 on B4141 Kingsdown Road when a van travelling north west from the college entrance, turned left and then across the main road into the path of a motorcyclist travelling south west, displacing the rider who sustained fatal injuries.

4.87 There are no notable clusters of accidents on these roads.

Area 7– Other unclassified roads

4.88 This area covers all other roads of the study area that are not included in the sections above.

4.89 A total of 59 PIAs were reported at this location during the 5-year study period, equivalent to 11.8 PIAs per annum. No fatal accidents were reported but eleven resulted in serious injury, eight of these occurring between 8pm and 3am.

4.90 Other notable factors amongst the 59 reported accidents are that 51% of them involved vulnerable road users: nine pedestrians, 12 pedal cycles and nine motorcycles. This high percentage of accidents involving vulnerable road users is indicative of the mixed use of these roads by different types of road user, for a multitude of local and longer distance activities.

4.91 There are no notable clusters of accidents on these roads.

Accident Summary

4.92 420 personal injury accidents were reported to have occurred within the study area during the most recent 5-year period for which information is available at the time of writing. This included two fatal accidents, both involving riders of two-wheeled vehicles. Overall there does appear to be a quite high proportion of rear shunt type accidents, and in several cases, accidents involving vulnerable road users, although this is not necessarily atypical of urban roads serving a mixture of purposes in a living environment.

4.93 The area has been split into several sections for further analysis and the numbers of accidents reported has been compared with statistics indicating expected rates for each type of road or junction given observed traffic levels. Overall the accident record appears to be quite good with observed figures generally below the equivalent expected rate.

4.94 The one notable exception to this is at White Hart Roundabout where a disproportionate number of rear shunt type accidents have occurred on three of the approaches to the junction. Proposals outlined elsewhere in this report to improve the capacity of this junction are likely to largely address this problem in the future by introducing signal control on two of the entries. These proposals will formalise the operation of junction through the introduction of traffic signals and therefore remove the decisions associated with drivers having to give way, simplifying their thought processes and thereby reducing errors of judgement.

4.95 To summarise, there are no notable safety issues that will remain unaddressed by the current development proposals. Although the development will add traffic to the network there is no reason to suppose that his will significantly compromise the relatively safe performance of the existing road system.

5 Development Proposals

Development Proposals

5.1 Full details of the development proposals are outlined in Section 2 of this document.

Development Timescales

5.2 The timescales for development delivery are dependant on many factors, including the planning process and future market demand for housing. This quantum of development will be constructed over a number of years, potentially circa 10 - 15 years. Therefore the development is likely to be built in several phases with points of access delivered progressively, as necessary.

Transport Strategy

5.3 To create a sustainable development it is fundamental to the process that the assessment considers how the future residents will access the Eastern Villages through all modes of transport. A sequential approach is to be followed, as detailed below:

- Encouraging environmental sustainability: Reducing the need to travel, especially by car
- Managing the existing network: Making best possible use of existing transport infrastructure
- Mitigating residual impacts: Initially through improvements to the local public transport network, and walking and cycling facilities, and then through provision of new or expanded roads.

Development Phasing and Access Strategy

5.4 As the development will be progressed over some time, it will be necessary to phase infrastructure delivery to meet the requirements of the growing community. The development will be built in several phases, with points of access onto the existing highway network being delivered as necessary. The principal points of access are indicated illustratively below and shown in greater detail in the later sections of this report.

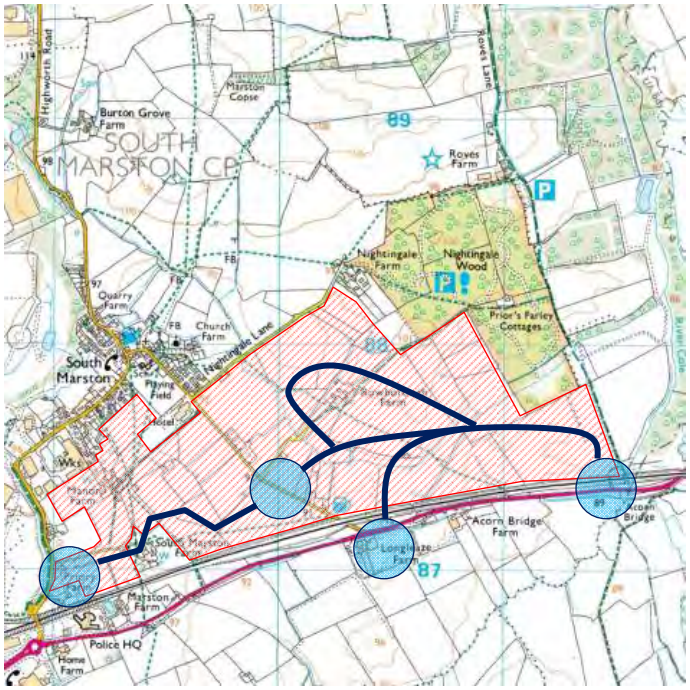

Figure 5a: Phase 1 access junction locations

5.5 The following paragraphs identify a series of indicative infrastructure phases that are proposed to support the proposed development.

Phase 1 A

5.6 At the time of writing, it is envisaged that the development will commence from the western boundary of the site with access taken from Thornhill Road via a new compact roundabout. It is envisaged that this could serve the first 200 – 300 dwellings at South Marston.

Phase 1B

5.7 A secondary access point will be delivered to the western side of Old Vicarage Lane via a three arm roundabout. In additional to this, the horizontal alignment of Old Vicarage Lane will be amended to provide a further access into the site. This will result in Old Vicarage Lane giving way to the main line flow which will be towards the development. This will have the benefit of reducing any desire line towards South Marston.

5.8 As development progresses, the connection of the access points onto Old Vicarage Lane and Thornhill Road will allow completion of the 580 dwellings identified to the south of South Marston by providing a second access point for each new home in the expanded South Marston.

Phase 1C

5.9 Access to the Rowborough phases of development will be taken from the eastern side of Old Vicarage Lane via a roundabout identified as part of Phase 1B, and as described in the work completed by JMP to support the South Marston SPD, together with a new priority junction further to the south. The horizontal alignment of Old Vicarage Lane will be amended to result in Old Vicarage Lane to the north giving way to the main line flow, which will be towards the development.

5.10 Access to Rowborough is also to be delivered to the east of the development, on the A420, via a signal controlled priority junction. The phasing of the connections back to the A420, including the timing of the delivery of the rail under bridge at the eastern end of the site, will be a matter for discussion with SBC.

5.11 The access strategy drawings are attached in Appendix C and show illustratively below.

Figure 5b: Development wide access strategy

5.12 The access strategy together with the internal road layout has been developed specifically to minimise the traffic routing through the proposed development and the existing South Marston village. The intention is to maintain the A420 as the primary east-west corridor and providing direct and ready connection to the A420 at Thornhill Road, Old Vicarage Lane, and from the new access towards the eastern end of the development The main connections through the site have been designed as public transport routes providing effective public transport connectivity between South Marston and Rowborough to enhance the attractiveness of public transport for residents of the new development but also the existing community. The link between Thornhill Road and Old Vicarage Lane has been designed with Manual for Streets philosophy in mind and will be experienced as a street within the landscape setting or context of South Marston rather than as a highly engineered road.

5.13 If not carefully designed, Old Vicarage Lane could result in additional traffic through South Marston. To reduce the attractiveness of this route, the access strategy reduces the importance of this link by altering the priorities with the development routes. This will encourage trips direct into the development rather than South Marston as was envisaged by the JMP background studies. By providing the second access to the Old Vicarage Lane, traffic generated from within Rowborough towards Old Vicarage Lane will employ the access(es) to the A420 to the south rather than find its way to Thornhill Road and Gablecross roundabout via Thornhill Road.

5.14 Consideration has also been given to the wider objectives in relation to transport in South Marston as set out in the Draft SPD. The objective of a 20 MPH speed limit throughout the village is supported in principle and in practice through the design of most streets within the development areas in accordance with such speeds. Rowborough and Nightingale Lanes are retained. The aspiration in Policy RA3 of the emerging Local Plan to secure traffic management measures in the expanded and existing village (for instance at Pound Corner) are also capable of being supported through the application proposals.

Internal Highway Network

5.15 Within the site, the masterplan proposes a street network having a clear hierarchy. The masterplan is indicative and will be confirmed through Reserved Matters. This is illustrated in Figure 5a and descried below:

Primary route: A main streets through South Marston and through Rowborough will connect to the external access points. The design speed for the internal street is based on a speed limit of 30mph, although the aspiration of the development is to achieve lower speeds through careful design of the streetscape and public realm. The main links through the development will be designed to cater for public transport vehicles..

It is envisaged that pedestrian and cycle movements will be catered for through on and off carriageway provision.

Secondary Routes: Secondary routes are designed to penetrate the individual development blocks and cater for vehicles at the reduced speeds, which will be reflected in the design and appearance of these roads.

Tertiary Routes: These will be designed to penetrate individual housing clusters and will be designed to encourage lower vehicle speeds and could incorporate shared spaces between motor vehicles, pedestrians and cyclists. The aspiration is for design speeds of 20 MPH on tertiary and secondary routes, thereby affording priority to walking and on street cycle movements as well as enhancing the public realm.

Walking and Cycling Provision

5.16 Published good practice identifies five main requirements for pedestrian routes. Wherever possible these should be followed when planning for pedestrians within the proposed development:

- Convenience – follow desire lines without any undue deviation from route,
- Connectivity – link multiple origin and destinations,
- Conviviality – be pleasant to use,
- Coherence – be made legible through paving and/or signage,
- Conspicuousness – promote security and safety allowing pedestrians to see and be seen by others

5.17 The ‘Guidance for Cycle Audit and Cycle Review’ (The Institution of Highways and Transportation, 1998) determines five main requirements for cycle routes. It is highly crucial that these requirements are recognised if the promotion of cycling to the site as a viable and attractive alternative to car use is to be successful:

- Coherence: continuous and to a consistent standard,
- Directness: closely follow desire lines as much as possible,
- Attractiveness: in aesthetic as well as objective terms
- Safety: designed to minimise risks for cyclists and others; and
- Comfort: well maintained smooth dry surfaces, flush kerbs and gentle gradients

5.18 Overall consideration should be given towards the former Commission for Architecture and the Built Environment (CABE) principles of inclusive design, as highlighted below:

- Inclusive: so everyone can use it safely, easily and with dignity.
- Responsive: taking account of what people say they need and want.
- Flexible: so different people can use them in different ways.
- Convenient: so everyone can use them without too much effort or separation.
- Accommodating: for all people, regardless of their age, gender, mobility, ethnicity or circumstances.
- Welcoming: with no disabling barriers that night exclude some people.
- Realistic: offering more than one solution to help balance everyone's needs and recognising that one solution may not work for all

5.19 The strongest desire line for walking and cycling is westbound. The distribution of trips, as indicated later in this report is 74% towards Swindon. The established peak hour car based trips of 788 can be used to determine the likely pedestrian / cyclist demands using census data. The 788 car based trips divided by the modal split of 0.695 provides a total of 1,134 trips. Pedestrian / cyclist trips represent 0.097 of the total trips, equating to 110 walking and cycling trip in the peak hour. This demand demonstrates the need for a strong walking and cycling network, which has been incorporated into the proposals.

5.20 The masterplan for the site will include numerous walking and cycling routes within the development to provide a comprehensive route network that will comprise both on and off road paths. Highway crossing points will be designed to cater for all types of pedestrian users with the routes lit where appropriate.

5.21 The walking and cycling paths will connect the individual housing blocks into the main route through the site that will ensure full connectively and route choice throughout the development.

5.22 The onsite network will connect into the external walking and cycling networks that includes the PRoW network. The predominant walking and cycling desire line towards Swindon is to be fully incorporated into the links from the development. Currently, there is a high quality walking and cycling link adjacent to the A420 commencing from the Police headquarters junction heading in a westerly direction through the Gablecross junction to connect with the Whitehart interchange. The development will provide a connection into the walking and cycling link adjacent to Thornhill Road, which in turn links with the A420. This will ensure that trips into the supermarket and into Swindon can be carried out. The walking and cycling strategy plan is attached in Appendix D.

Road Based Public Transport Provision

5.23 To maximise the opportunities to travel by public transport, it is proposed to improve the current routes that operate in Swindon. The options to deliver public transport enhancements have been discussed with SBC and proposals delivered following various discussions with the local bus operator, Stagecoach.

5.24 Presently, two routes operated by Stagecoach pass adjacent to the site boundary. It has been agreed that the existing routes will be augmented and diverted into the proposed development.

5.25 The development is likely to commence at Thornhill Road and proceed broadly in an easterly direction. Therefore a connection through the development into Old Vicarage Lane is unlikely to be delivered at the outset. Through discussions with Stagecoach, the existing Route 66 can only practically be routed through the development once connection with Old Vicarage Lane is in place. Alternative routes have therefore been investigated to serve the in initial phases of the development.

5.26 Route 72 has been identified as a viable alternative to support the early phases. This route currently serves Wrougthon to the south of Swindon and terminates at the bus station. This route can be extended to serve the EV Monday to Saturday

at half hourly intervals with the first call at 06:24 with the last stop at 20:54. This provision will continue until a connection between Thornhill Road and Old Vicarage lane is established.

5.27 The overall public transport strategy is attached in Appendix E.

5.28 Once the link between Thornhill Road and Old Vicarage Lane is in place, Route 66 will be diverted into the site and Route 72 will revert back to the original provision, unless patronage is sufficient that the route is self funding.

5.29 It is proposed that Route 66 will operate at an improved 20 minute frequency Monday to Saturday and hourly on Sunday.

5.30 The long term viability of any public transport routes is critical if it is serve the community into the future. Therefore discussions with Stagecoach has established the likely level of revenue that could be generated by the development, which can be offset by the likely costs to understand the viability. As a result of this work, financial support is likely to be needed initially, but over time it is expected that patronage levels will be sufficient to safeguard the long term viability of the proposed public transport interventions.

Offsite Interventions

5.31 Through historical modelling of the transport impacts, various off site junctions were provisionally identified as requiring mitigation. Further assessments have been completed to support the production of this Transportation Assessment, which supports and refines the requirements for off-site infrastructure interventions. These are identified below:

- **A420 / Old Vicarage Lane:** Shuttle signals to operate at the railway underbridge on Old Vicarage Lane linked to a signal controlled junction between Old Vicarage Lane and A420, together with a change of priorities leading into the development on the eastern side of Old Vicarage Lane with Old Vicarage Lane becoming the minor road. The layout of this arrangement is shown on drawing 1335-HL-057 and 1335-HL-054 in Appendix C

- **New Access onto A420:** New signal controlled junction at the east end of Phase I to provide access from the A420. The layout of this arrangement is shown on drawing 1335-HL-10 in Appendix C

- **Police HQ Signals:** Retain the existing layout on A420, with minor changes to the signal timings. Consider an option to open the secondary exit point onto Thornhill Road, should the Police desire.

- **Gablecross** Roundabout: Localised widening on approach arms. The layout of this arrangement is shown on drawing 1335-HL-057 in Appendix C

- **White Hart interchange:** Partial signalisation of the eastern side of the interchange. The layout of this arrangement is shown on drawing 1335-HL-050 in Appendix C

This represents a comprehensive package of interventions delivered through, and arising from, the development. The detailed traffic modelling completed for this TA has also considered future development resulting from the wider growth proposals to the east of Swindon, as identified in the Local Plan. As would be expected, this work has identified further interventions that will be necessary to deliver the wider growth. However, the assessments demonstrates that delivery of this Phase I element of the growth does not prejudice the delivery of the subsequent phases. Such further interventions that have been anticipated in the Eastern Villages SPD for instance and include for diamond interchange at White Hart and the signalisation of the Greenbridge roundabout. The SPD envisages the phased implementation of wider transport improvements in accordance with the phased implementation of the eastern villages development.

Parking Provision

5.32 SBC has published guidance on parking standards, in which dwellings having four or less bedrooms shall have two spaces with all other dwellings having three spaces. This guidance was before the publication of NPPF, which at Paragraph 39 identifies that local planning authorities should take into account:

- Accessibility of the development
- The type, mix and use of development
- The availability of and opportunities for public transport
- Local car ownership levels
- An overall need to reduce the use of high-emission vehicles

5.33 The application is submitted in outline and parking numbers can be refined / defined at reserved matters stage.

5.34 It is anticipated that every dwelling will have access to safe, secure cycle parking. It is assumed that garages, where provided, will provide suitable cycling parking and for the dwellings without garages, it is anticipated that secure facilities will be provided either within the building or in rear gardens.

6 Development Impact Appraisal

Impact Appraisal

6.1 The DfT Guidance on Transport Assessment of March 2007 requires TA's to consider the impact of new development using the principles set out in the New Approach to Appraisal (NATA). The impact of proposals are assessed in terms of the five NATA objectives for transport:

- Accessibility
- Safety
- Economy
- Environment
- Integration

Accessibility

6.2 The accessibility of the development is achieved through the successful forming of transport links from the development to the external transport routes such that a permeable layout is delivered that allows the future site occupiers to access the current range of local facilities and amenities by different modes of travel.

6.3 A qualitative review of the accessibility implications of the proposed development has been conducted. The existing level of access for cyclists and pedestrians between the proposed development and the surrounding transport system is described in Chapter 4.

Employment Facilities

6.4 Various employment opportunities are located in close proximity to the site, including South Marston Park, that comprises a site area of 93.35 Ha and Keypoint at South Marston, comprising a site of 24.54Ha. Further afield, other key employment destinations include the Great Western Hospital, Green Bridge Retail Park, Swindon Town Centre together with a range of industrial / trading estates. These offer a wide range of employment opportunities for the future residents.

6.5 The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6a, with the location indicated in Figure 6b.

Employment	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. South Marston Park	0.5km	✓	6	✓	2
2. Keypoint	0.5km	✓	6	✓	2
3. Great Western Hospital	5.7km		68		20
4. St Margaret's Retail Park	0.4km	✓	5	✓	1
5. Green Bridge Retail Park	2.4km		29	✓	8
6. Swindon Town Centre	5.0km		60	✓	18

Figure 6a: Distance to Employment Destinations

Figure 6b: Employment Destinations

Health Facilities

6.6 A range of health facilities are available within Swindon. The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6c, with the location indicated in Figure 6d.

Health	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
1. Elm Tree Surgery	6.0km		72		21

2. Eldene Surgery & Pharmacy	5.4km	65		19
3. Penhill Surgery	5.9km	71		21
4. Hawthorn Medical centre	5.0km	60	✓	18
5. Kingswood Surgery	5.3lm	64		19
6. Park South Dental Practice	5.6km	67		20
7. Great Western Hospital	5.7km	68		20

Figure 6c: Distance to Healthcare Destinations

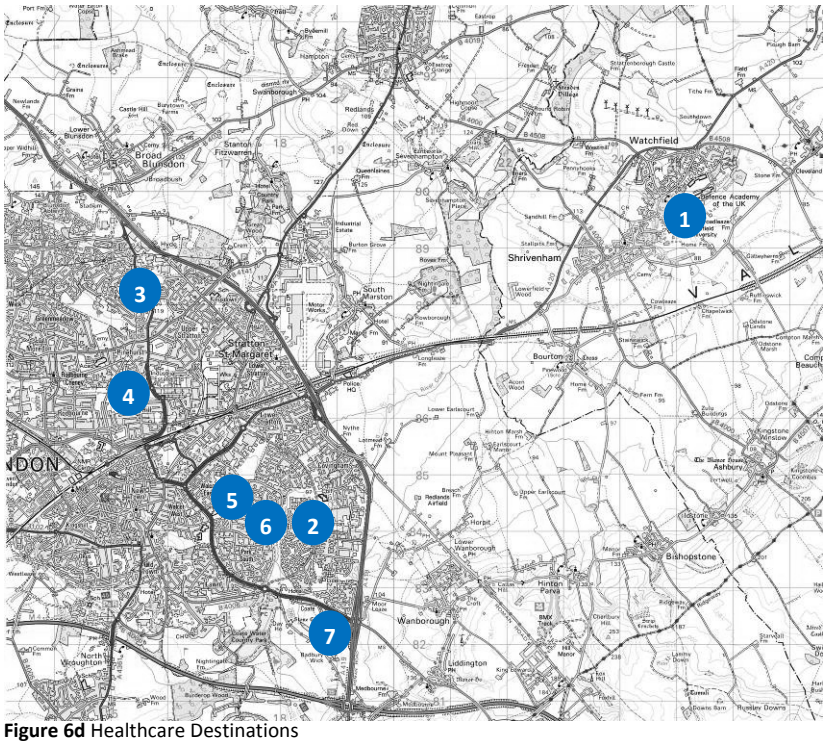

Figure 6d Healthcare Destinations

Education Facilities

6.7 There are a range of education facilities within Swindon. The distance to the key destinations, measured from the site access on Thornhill Road, is indicated in Figure 6e, with the location indicated in Figure 6f.

Schools	Approx Distance from proposed Site entrance (km)	Meet 2km Target Walk?	Approx Walk Time (mins)	Meet 5km Target Cycle?	Approx Cycle Time (mins)
Primary					